

Pass-through Beyond the Border: Oligopoly, Networks, and Tariffs*

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Abstract

We develop a multisector open-economy framework that maps border price shocks into US producer and consumer prices through imported inputs, domestic production networks, and competition with foreign suppliers. The framework accommodates pass-through in either levels or logs and maps directly to public data on input-output linkages, trade, and prices. We apply it to the 2025 US tariffs. At the precision available in industry-level price data, the level and log specifications are difficult to distinguish. In contrast, the role of market structure is clear: under either specification, we reject the monopolistic-competition benchmark in which domestic prices respond only to marginal costs. Allowing firms to respond to the prices of competing imported varieties substantially improves the model’s fit to producer prices, and the resulting model is also consistent with the consumer-price response. Aggregating the estimated price effects across the full consumption basket, the January–October 2025 tariff increases imply a first-order contribution of 0.93 percentage points to headline PCE inflation, including 0.11 percentage points from the foreign-competition channel.

Keywords: Tariffs, inflation, import prices, indirect imports
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1 Introduction

When a tariff is imposed on a product from a country at the border, its downstream effect on US prices depends on how imported goods enter domestic production, how cost shocks move through input-output linkages, how retailers react to higher finished-goods prices, and how domestic firms set prices when they compete with foreign suppliers. Because these forces interact, the literature has reached broad agreement on pass-through at the border and much less agreement on pass-through into domestic producer and consumer prices.

We build a multisector open-economy model with oligopolistically competitive retailers and intermediate-goods producers. The model can account for pass-through of marginal costs in levels, network amplification, and competition effects generated by a tariffs. There are two main contributions from this setting. First, the richness of our model helps organize a literature that has produced relatively consistent evidence on border pass-through but more dispersed estimates of pass-through into domestic producer and consumer prices. Some of the disagreement across studies can be reconciled by different assumptions and by conditioning on different channels. This is our attempt to spell out some canonical channels that can generate pass-through heterogeneity in the context of industry-level pass-through estimation of foreign price shocks. Second, the model allows us to map with precision every dollar paid at the border to implied costs for domestic producers' and consumers that span virtually all Producer Price Index and Personal Expenditure index categories. This allows not only to exploit a wide set of observations for identification but also to account for the full (partial-equilibrium) aggregate effects of tariffs in the economy.

The model is based on an assignment problem with discrete firm choice, which delivers pass-through in levels and allows us to easily compare it with full pass-through in logs. Firms and retailers choose prices under Bertrand competition, thus allowing foreign prices to affect domestic companies both via marginal costs and via their reference price. Firms and retailers are also heterogeneous in terms of marginal cost exposure to foreign price shocks, thus potentially creating incomplete pass-through according to the interaction of a firm's exposure to the shock and its pass-through. We derive a first-order mapping from foreign price shocks to domestic producer prices that depends on three objects: a markup/pass-through term, an oligopolistic competition term, and a network amplification term of both terms. A more general version of this paper with terms summarizing firm-level correlations between marginal cost exposure and firm level pass-through is also derived in the Appendix.

Taking that mapping to the data requires a measurement chain that we regard as a contribution in its own right, because the classification systems it spans have no published concordance linking them end to end. We move from the legal tariff schedule at the HS-10-by-

country-of-origin level, to duties actually collected at the border, to BEA Detail commodities through the regional Make matrix, to 273 PPI series, and then through the PCE bridge to NIPA expenditure categories, carrying the distribution margin that separates producer from purchaser prices. The same chain delivers the final-import component of exposure, through imported finished goods that consumers buy, and the domestic-production component, through imported inputs propagated recursively across the domestic network. Both are built from BEA input-output accounts, Census trade records, and BLS price indices only.

The first link in that chain is testable on its own. At the level of individual import records we find that duties pass into duty-inclusive border prices essentially one for one, 0.93 on impact and 0.99 by the third month, with no sign that foreign exporters cut their declared prices. We can thus treat the observed tariff change as the model’s foreign price shock in every step that follows.

We estimate the price response horizon by horizon, using local projections (Jordà, 2005) in a difference-in-differences design following Dube et al. (2025), with item and date fixed effects so that identification comes from cross-sectional differences in model-implied exposure. We use statutory tariff changes as instrument. The design conditions on the forward path of tariffs, so the estimand is the response to a permanent tariff increase. Conditioning in this way allows the horizon at which the coefficient reaches one to be read as the speed of adjustment towards the flexible-price response, consistent with the model.

Because the regressor is a predicted price change rather than a tariff rate, its coefficient is a test of the model: a value of one says that observed prices moved by the amount the model predicts. Our central result is a sequence of tests where a model-implied relative price response is compared to observed relative price response. On producer prices in 2025, a version of the model in which a tariff reaches domestic firms only through their own costs delivers a model pass-through of 2.946 (s.e. = 0.672), meaning that observed price variation moves three times as much as model implied prices. Adding propagation through the domestic input-output network delivers 2.314 (s.e. = 0.537). Allowing the same tariff to raise the price of the imported varieties domestic firms compete against delivers 1.173 (s.e. = 0.271). Carrying that estimate to consumer prices delivers a pass-through of 0.929 (s.e. = 0.245). In short, we find evidence of full pass-through of our model-implied price increases once accounting for network amplification and competition effects.

We discipline the competition channel above by estimating a common conditional pass-through parameter $\bar{\rho}$ using a generalized methods of moments (GMM) estimator. The estimator requires the model to match both the scale of the producer-price long-term response and its cross-sectional shape. That is, in addition to requiring predicted price pressure to move observed prices one for one, we use the model-implied change in exposure with respect

to $\bar{\rho}$ to identify how tariff pressure is redistributed across industries as the competition channel becomes stronger. During the 2025 tariff episode, this yields $\hat{\rho} = 0.655$, with a 95% confidence set of $[0.433, 0.886]$. We reject that $\bar{\rho} = 1$, which supports the existence of strategic complementarities observable from industry data. The estimate is also robust to allowing for within-industry covariance between firms' import exposure and their pass-through: estimating that wedge jointly gives $\hat{\rho} = 0.719$. The 2018–19 tariff episode provides similar results, $\hat{\rho} = 0.697$, but is considerably less informative: the specification exhibits pre-trends and concentrated identifying variation, and the corresponding confidence set $[0.238, 1.000]$ contains both the monopolistically-competitive benchmark and our 2025 estimate.

Because the model maps the tariff schedule into the full PCE expenditure system, the same category-level objects used to estimate pass-through also aggregate into a first-order contribution to aggregate PCE inflation when weighted using observed expenditure weights. Cumulated over the 2025 window, the three channels sum to 0.93 percentage points of headline PCE, of which the competition increment contributes 0.11. Although this exercise is first-order partial-equilibrium, its advantage is that it only requires knowledge of one estimated production-side parameter and observed weights.

We can further decompose the aggregate contribution to headline PCE. Splitting the estimate by month identifies which tariff waves account for the price pressure, while decompositions by tariffed product and source country attribute the effect to the parts of the schedule that generated it. A separate decomposition by PCE category instead shows where those shocks ultimately appear in consumer expenditure. The distinction is useful because the products and countries on which tariffs are imposed need not coincide with the consumption categories in which their effects are ultimately recorded once imported inputs, production networks, and distribution margins are taken into account. The largest contributions come from the expected categories and origins: new motor vehicles, garments, furniture, and consumer electronics are the largest consumption categories, and China accounts for two fifths of the total. But a quarter of the contribution comes from services, which contain no imported final goods and are reached only through the domestic production network, and from categories each below one percent of the total. Half of the contribution attributed to Canada, Brazil, and Germany also arrives through input costs rather than as final imports, against a fifth for Vietnam.

Our contribution is therefore threefold. First, we provide a unified mapping from the legal tariff schedule to border prices into the full consumption basket accounting for both domestic production networks and the presence of oligopolistic domestic pricing. Second, we estimate and test that mapping dynamically in producer and then consumer prices. Finally, we combine the estimated mapping together with observed expenditure weights to price the

contribution of a tariff schedule to aggregate PCE inflation.

Related Literature. Our paper contributes to three strands of research.

The first is the empirical literature on tariff pass-through. A large body of work studies the 2018–19 US tariff episode. [Amiti, Redding and Weinstein \(2019\)](#), [Fajgelbaum et al. \(2020\)](#), and [Cavallo et al. \(2021\)](#) find that the tariffs were largely passed through to US import prices at the border, with limited absorption by foreign exporters, and our border estimates reproduce that finding for 2025. A related set of papers studies how these border-price increases affected downstream domestic prices. [Cavallo et al. \(2021\)](#) trace tariff effects into retail prices for selected product categories, and [Flaaen and Pierce \(2024\)](#) estimate pass-through to producer prices in manufacturing. More recent work studies the 2025 episode. [Minton and Somale \(2025\)](#) and [Minton, Ray and Somale \(2026\)](#) develop real-time methods for detecting tariff effects in consumer prices using public data. [Cavallo, Llamas and Vazquez \(2025\)](#) use high-frequency retail microdata matched to product-level tariff rates and countries of origin to measure short-run retail-price effects. [Hacıoğlu-Hoke, Malladi and Feler \(2026\)](#) document the gradual retail-price response using item-level spending data and information on countries of production. [Amiti et al. \(2026\)](#) study the incidence of the 2025 tariffs at the border, and [Fajgelbaum and Khandelwal \(2026\)](#) and [Gopinath and Neiman \(2026\)](#) analyze the broader short-run effects and incidence of the recent tariff increases.

Relative to this literature, our contribution is to provide a framework that clarifies which downstream price response is being estimated. A tariff can affect consumer prices through imported final goods, through imported intermediate inputs, through domestic input-output propagation, and through the pricing response of domestic competitors. Empirical designs that include different subsets of these channels need not estimate the same object. Our framework maps tariff shocks from product-country cells to producer industries and PCE expenditure categories, separates final-import from domestic-production exposure, and then asks which additional pricing channels are needed to reconcile the model-implied price response with observed PPI and PCE price changes.

The consumer-price mapping builds on [Barbiero and Stein \(2025\)](#), which introduced an input-output-based mapping from border-price shocks to the PCE consumption basket, alongside related early work by [Minton and Somale \(2025\)](#). We extend that approach from an accounting exercise to a framework in which the same mapping is used to identify pass-through, production-network propagation, and oligopolistic responses to observed tariff shocks.

Subsequent work has developed complementary approaches to measuring the consumer-price effects of the 2025 tariffs. [Minton, Ray and Somale \(2026\)](#) extend the real-time PCE

approach of [Minton and Somale \(2025\)](#), while [Amiti, Heise and Weinstein \(2026\)](#) estimate direct-import, imported-input, and strategic-complementarity channels using matched import, producer-price, and CPI data. That paper is the closest to ours in the set of channels it considers, and our object and empirical design differ from it in several respects. We propagate imported-input shocks through the domestic input-output network. We estimate dynamic pass-through with a local-projection difference-in-differences design and we estimate a model-consistent parameter that regulates marginal cost and competitors’ price pass-through via GMM. Our PCE mapping spans the consumption basket, allowing the cross-sectional categories used for identification to aggregate directly to an economy-wide first-order contribution to PCE inflation. This distinction matters for aggregation: because the model maps the tariff schedule into the full PCE expenditure system, the same category-level objects used to estimate pass-through also aggregate, with observed expenditure weights, into a first-order contribution to aggregate PCE inflation.

The second strand is the literature on production networks, inflation, and shock propagation. [Baqae and Farhi \(2019\)](#) and [Baqae and Rubbo \(2023\)](#) develop general-equilibrium frameworks for studying how shocks propagate through input-output linkages and affect aggregate outcomes. [La’O and Tahbaz-Salehi \(2022\)](#) study optimal monetary policy in production networks. Related work emphasizes that supply-chain shocks can affect inflation with delays and through heterogeneous downstream exposure; [Minton and Wheaton \(2023\)](#), for example, document delayed inflation in supply chains. We build on the first-order network approximation logic in this literature, but focus on a partial-equilibrium object that is directly useful for tariff and import-price scenarios: the mapping from a vector of border price shocks to predicted PPI and PCE price changes, accounting for oligopolistic competition and heterogeneous firm exposure. We also show how to construct that mapping from public US national-accounts data, including BEA input-output tables, import-use matrices, the Make table, and the PCE bridge.

The third strand is the literature on markup adjustment, imperfect competition, and pass-through. Classic contributions such as [Feenstra \(1989\)](#) and [Goldberg and Knetter \(1997\)](#) show that pass-through is generally incomplete under imperfect competition. [Atkeson and Burstein \(2008\)](#) emphasize strategic complementarities in oligopolistic markets, where firms’ price responses depend on the prices of their competitors. More recently, [Sangani \(2026\)](#) argues that much of the evidence that appears to be incomplete pass-through in percentages is consistent with complete pass-through in levels. That distinction is central here: under pass-through in levels, a cost increase raises prices dollar for dollar, so percentage pass-through is attenuated by the cost share of price. We find that accounting for pass-through in levels is not by itself enough to reconcile the model with the producer-price response, which is why

the foreign-competition channel is estimated rather than assumed away.

The closest paper in this strand is [Amiti, Itskhoki and Konings \(2019\)](#), who show that aggregate pass-through depends not only on average exposure to international shocks but also on the covariance between firm-level exposure and markup adjustment. We build on that insight and change the object of measurement, while also adding network effects to that framework. Rather than estimating exchange-rate pass-through into a sectoral price index, we condition on post-border foreign price changes and trace their effect through domestic producer and consumer prices, which separates two forces that a sectoral exchange-rate exercise combines: foreign price shocks reach domestic firms through imported-input costs, and they also reach them by raising the prices of foreign competitors. We embed both forces in an input-output system. In the appendix extension, we also allow the domestic price response to depend on the covariance between firms' exposure to the relevant shock and their pass-through.

2 Model

Notation. There is a set $\mathcal{I}$ of industries. There is also a set $\mathcal{E}$ of expenditure categories.

We use i and j , and occasionally k , to index industries. We use e to index expenditure categories.

Each industry i contains a discrete number of firms $\mathcal{N}_i$. The number of firms comprises a set of domestic firms $\mathcal{D}_i$ and a single foreign firm F_i , so that $|\mathcal{N}_i| = |\mathcal{D}_i| + 1$. We use f, g, h to index firms.

Each expenditure category e also contains a discrete number of firms $\mathcal{N}_e$. In contrast to industries, all firms within expenditure categories are domestic. As in the industry case, we use f, g, h to index firms within expenditure categories as well.

2.1 Intermediate Input Producers

2.1.1 Procurement

To generate pass-through in levels in intermediate-input markets, we model a procurement assignment problem for an industry entity that requires a continuum of varieties to be completed by a discrete number of firms to produce an aggregate good. This aggregated good serves as a reference for the oligopolistic competition setup that follows. Our setup thus belongs to the class of *shift-invariance* models analyzed in [Sangani \(2026\)](#). Here, we provide a microfoundation for the possibility of incomplete pass-through in levels within intermediate-input markets.

The problem of the industry entity is to choose what varieties to assign to firms. Each variety is assigned to a single firm, but firms may complete more than one variety. Thus, for each variety, this is a discrete choice problem of choosing the firm that produces it.

A well-known result in the discrete choice literature is that this problem can be written either as a standard discrete choice maximization problem or as an expected minimization problem with additional entropy-regularized cost. To ease exposition, we proceed with the second route. For additional details on the exact mathematical foundation of this duality, see Section 1.3 in [Galichon \(2026\)](#).

The industry entity i chooses variety shares to allocate to each firm f (π_{fi}) that minimize an expected cost problem with an additional entropy-regularized procurement cost as follows:

$$p_i^{eff} = \min_{\pi_{fi} \in \Delta_i} \left\{ b_i + \sum_f \pi_{fi}(p_{fi} - z_{fi}) + \sigma_i \sum_f \pi_{fi} \log \pi_{fi} \right\}, \quad (1)$$

where $\Delta_i = \{\pi_{fi} : \sum_f \pi_{fi} = 1\}$ is the simplex set.

p_i^{eff} represents the *effective price index* of industry i . b_i captures a baseline cost of satisfying an industry- i variety. An example is a common physical or administrative cost of completing an industry i requirement: handling, distribution, assembly, etc. It is common across all alternatives. Its sole role is to pin down the level of the effective price index, as defined below, and ensure that it is positive. In contrast, z_{fi} is a firm-specific *deterministic* advantage in serving industry- i varieties, measured in money units. We can think of it as quality, reliability, and delivery speed, among others. σ_i represents the elasticity of substitution across varieties/firms within an industry.

The solution to the entropy-regularized problem delivers *variety-shares* π_{fi}

$$\pi_{fi} = \frac{\exp\left(\frac{z_{fi} - p_{fi}}{\sigma_i}\right)}{\sum_{g \in i} \exp\left(\frac{z_{gi} - p_{gi}}{\sigma_i}\right)} \quad (2)$$

Importantly, π_{fi} is not a quantity; it is an assignment share. It tells us the share of industry i varieties, orders, or procurements assigned to firm f .

Replacing equation (2) into the effective price problem delivers the *effective cost index of industry i* as

$$p_i^{eff} = b_i - \sigma_i \log \left[\sum_{f \in i} \exp\left(\frac{z_{fi} - p_{fi}}{\sigma_i}\right) \right]. \quad (3)$$

The effective cost index depends on z_{fi} and b_i as well as on p_{fi} . As a result, it is more akin to a *quality-adjusted price* than to a standard price index. Using this definition, we can rewrite the assignment share as

$$\pi_{fi} = \exp \left(\frac{z_{fi} - b_i + p_i^{eff} - p_{fi}}{\sigma_i} \right) \quad (4)$$

The quantity produced by firm f in industry i is:

$$q_{fi} = \pi_{fi} q_i, \quad (5)$$

where q_i is the real composite output of industry i from the supply-side.

Sales/Revenues of firm f in industry i are then

$$R_{fi} = p_{fi} q_{fi} = p_{fi} \pi_{fi} q_i. \quad (6)$$

Remark 1. *Under the logit structure, we have*

$$p_i^{eff} q_i \neq \sum_{f \in i} p_{fi} q_{fi}. \quad (7)$$

Remark 1 highlights that we cannot separate a quantity and a price index as we can under constant elasticity of substitution (CES) aggregation.

Using the definition of sales, we can construct sales shares for each firm in the industry (s_{fi}) as :

$$s_{fi} = \frac{R_{fi}}{\sum_{g \in i} R_{gi}} = \frac{p_{fi} \pi_{fi}}{\sum_{g \in i} p_{gi} \pi_{gi}} = \frac{p_{fi} \exp \left(\frac{z_{fi} - p_{fi}}{\sigma_i} \right)}{\sum_{g \in i} p_{gi} \exp \left(\frac{z_{gi} - p_{gi}}{\sigma_i} \right)}. \quad (8)$$

Remark 2. *Under the logit structure, sales shares are no longer homogeneous of degree 0 in prices.*

Remark 2 highlights a key property of shift-invariant models: sales shares are no longer homogeneous of degree zero in firms' prices within the industry.

For future reference, it will prove useful to define the sales share of the foreign firm (s_{Fi}) as a separate entity:

$$\sum_{f \in \mathcal{D}_i} s_{fi} + s_{Fi} = 1. \quad (9)$$

2.1.2 Within Industry Competition

Each domestic firm $f \in \mathcal{D}_i$ solves the following program:

$$\Pi_{fi} = \max_{p_{fi}} (p_{fi} - mc_{fi})\pi_{fi}q_i \quad \text{s.t.} \quad (2). \quad (10)$$

Note that we assume Bertrand competition in what follows, since firms choose prices directly.

At this point, we need to make the following assumption:

Assumption 1. *Firms consider their effect on the variety share π_{fi} , but not the impact of their pricing decisions on the inputs they source.*

Assumption 1 rules out input market power and allows us to focus on the role of competition across firms. This is a strong assumption that we are forced to make given our data availability. The implication of this assumption is that we can take firms' marginal costs as given when making the pricing decision.

The solution to this problem delivers the pricing condition:

$$p_{fi} = mc_{fi} + \mu_{fi}, \quad (11)$$

where $\mu_{fi} = \frac{\sigma_i}{1-\pi_{fi}}$ is firm f 's markup. Importantly, the price level p_{fi} is linear in the markup and the marginal cost (μ_{fi}, mc_{fi}) . Markups can react in *levels* in our setup, so incomplete pass-through in levels is a possibility.

The production function of firms is constant returns to scale and takes the following form

$$q_{fi} = q_{fi}(\ell_{fi}, x_{fi}, x_{fi}^*; a_{fi}), \quad (12)$$

where ℓ_{fi} is labor demand, x_{fi} is a domestic intermediate input bundle, x_{fi}^* is an imported intermediate input and a_{fi} is a firm-specific productivity.

The domestic intermediate input bundle is a composite of industry-specific bundles $x_{fi,j}$

$$x_{fi} = x_{fi}(\{x_{fi,j}\}_{j \in \mathcal{I}}). \quad (13)$$

The industry-specific bundles are themselves an aggregation across *domestic* firms within each industry

$$x_{fi,j} = x_{fi,j}(\{x_{fi,gj}\}_{g \in \mathcal{D}_j}). \quad (14)$$

The imported intermediate input bundle is a composite of industry-specific import bundles

$x_{fi,j}^*$:

$$x_{fi}^* = x_{fi}^*(\{x_{fi,j}^*\}_{j \in \mathcal{I}}). \quad (15)$$

The price of each industry-specific import bundles is p_j^* and is taken as given.

Given our setup, marginal costs satisfy

$$mc_{fi}(w, p_{fi}^x, p_{fi}^{x*}; a_{fi}), \quad (16)$$

where w is the wage, p_{fi}^x is the price of the domestic intermediate input bundle, p_{fi}^{x*} is the price of the imported intermediate input bundle and a_{fi} is firm's productivity. Note that for now we leave both prices of intermediate input bundles as firm-industry specific.

2.2 Final Demand

2.2.1 Assignment Problem

We model final consumers in a manner similar to that of input markets. We consider a discrete set of expenditure categories $e \in \mathcal{E}$. Within each expenditure category e there is a continuum of varieties $\omega \in [0, c_e]$. A discrete number of firms $\mathcal{N}_e$ produce these varieties. The household's problem is to decide the share of varieties to buy from each firm in each category. For each category e , the household problem is then:

$$p_e^{eff} = \min_{\pi_{fe} \in \Delta_e} \left\{ b_e + \sum_f \pi_{fe}(p_{fe} - z_{fe}) + \sigma_e \sum_f \pi_{fe} \log \pi_{fe} \right\}, \quad (17)$$

where $\Delta_e = \{\pi_{fe} : \sum_f \pi_{fe} = 1\}$ is the simplex set.

As before, the solution to this problem delivers:

$$\pi_{fe} = \frac{\exp\left(\frac{z_{fe} - p_{fe}}{\sigma_e}\right)}{\sum_{g \in e} \exp\left(\frac{z_{ge} - p_{ge}}{\sigma_e}\right)}, \quad (18)$$

which is the share of varieties that households buy from firm f in category e . The demand for a firm's output is thus

$$c_{fe} = \pi_{fe} c_e. \quad (19)$$

2.2.2 Retailers

We model the problem of firms within each category as a Bertrand competition. The problem of each retailer is thus

$$\Pi_{fe} = \max_{p_{fe}} (p_{fe} - m_{fe})\pi_{fe}c_e \quad \text{s.t.} \quad (18) \quad (20)$$

This delivers the pricing condition:

$$p_{fe} = mc_{fe} + \frac{\sigma_e}{1 - \pi_{fe}}. \quad (21)$$

The retailers production function is constant returns to scale

$$c_{fe} = c_{fe}(\ell_{fe}, \{x_{fe,gj}\}_{g \in \mathcal{D}_j, j \in \mathcal{I}}, \{x_{fe,j}^*\}_{gj \in \mathcal{I}; a_{fe}}), \quad (22)$$

where ℓ_{fe} is labor usage, $x_{fe,gj}$ is domestic intermediate input demand, $x_{fe,j}^*$ is imported intermediate input demand and a_{fe} is a productivity.

The production function delivers a marginal cost function for the retailers

$$mc_{fe} = mc_{fe}(w, \{p_{gj}\}_{g \in \mathcal{D}_j, j \in \mathcal{I}}, \{p_j^*\}_{j \in \mathcal{I}; a_{fe}}) \quad (23)$$

2.3 Market Clearing

Total sales of firm f in industry i must satisfies the following market clearing condition

$$p_{fi}q_{fi} = \sum_{j \in \mathcal{I}} \sum_{g \in \mathcal{D}_j} p_{fi}x_{gj,fi} + \sum_{e \in \mathcal{E}} \sum_{h \in e} p_{fi}x_{he,fi}, \quad (24)$$

where the right hand side is demand by intermediate producing firms plus demand by retailers.

Total labor payments of the household satisfies:

$$w\bar{L} = \sum_{j \in \mathcal{I}} \sum_{g \in \mathcal{D}_j} w\ell_{gj} + \sum_{e \in \mathcal{E}} \sum_{h \in e} w\ell_{he}. \quad (25)$$

2.4 Price Indices Definitions

2.4.1 Industry level price indices

We define the sales weighted average price index change of industry i as

$$d \log p_i = \sum_{f \in \mathcal{D}_i} s_{fi} d \log p_{fi} + s_{Fi} d \log p_{Fi}. \quad (26)$$

This index is observed in the data but it does not correspond to any market price in the usual sense.

The *observed* price index change of industry i , $d \log p_i$, is linked to the effective price change as

$$dp_i^{eff} = \frac{R_i}{q_i} d \log p_i \quad (27)$$

Define domestic sales as $R_i^d = \sum_{f \in \mathcal{D}_i} R_{fi}$ and domestic sales shares as

$$s_{fi}^d = \frac{R_{fi}}{R_i^d}. \quad (28)$$

Let the industry-level variable costs be

$$VC_i^d = \sum_{g \in \mathcal{D}_i} m_{c_{gi}} q_{gi}; \quad VC_i = \sum_{g \in i} m_{c_{gi}} q_{gi}, \quad (29)$$

where VC_i^d is variable costs across domestic firms and VC_i is the industry's total variable costs, inclusive of those of the foreign firm.

We also define the domestic price index of industry i as a domestic-sales weighted average across domestic firms $\mathcal{D}_i$

$$d \log p_i^d = \sum_{f \in \mathcal{D}_i} s_{fi}^d d \log p_{fi} \quad (30)$$

Equation (30) defines the producer price index (PPI) of industry i as in the data.

2.4.2 Consumer's price indices

We define the price index change in a category e as

$$d \log p_e = \sum_{f \in e} s_{fe} d \log p_{fe}, \quad (31)$$

where $s_{fe} = \frac{p_{fe} c_{fe}}{\sum_{g \in e} p_{ge} c_{ge}}$ is the expenditure share on firm f within category e .

Using this definition, we further define the consumer price index change as in the data:

$$d \log p^C = \sum_{e \in \mathcal{E}} s_e d \log p_e, \quad (32)$$

where $s_e = \frac{\sum_{g \in e} p_{ge} c_{ge}}{\sum_{e \in \mathcal{E}} \sum_{g \in e} p_{ge} c_{ge}}$ is the expenditure share in category e .

2.5 The Transmission of Import Prices

In this section, we derive the key equations that we bring to the data. To do so, we consider a first-order approximation around an initial equilibrium.

2.5.1 Pass-through

The change in the price of a firm f -industry i can be written as a function of both marginal cost changes and changes in competitors' prices. The following proposition characterizes this result in levels:

Proposition 1 (Pass-through in levels). *Up to a first-order approximation, the level change in the price of firm f in industry i can be written as*

$$dp_{fi} = (1 - \pi_{fi}) dmc_{fi} + \pi_{fi} dp_{-f,i}, \quad (33)$$

where

$$dp_{-f,i} = \sum_{h \in i \setminus f} \tilde{\pi}_{hi} dp_{hi} \quad \text{and} \quad \tilde{\pi}_{hi} = \frac{\pi_{hi}}{\sum_{g \in i \setminus f} \pi_{gi}}, \quad (34)$$

is the competitor's price index for firm f and $\tilde{\pi}_{hi}$ is the variety share of firm h in industry i within varieties performed by domestic firms.

Proof. See Appendix B.1. □

Proposition 1 shows that changes in the firm's price can be decomposed between an own marginal cost change and a competitors' price index. The complete pass-through in level is achieved whenever $\pi_{fi} = 0$ and corresponds to the case analyzed in Sangani (2026). In this case, the fraction of varieties a particular firm performs is negligible relative to the aggregate, an intuition similar to that in monopolistic competition models. By varying π_{fi} , we can nest different cases from the literature, but under pass-through in levels.

The next proposition rewrites the firm-level pass-through in percentage changes¹:

Proposition 2 (Pass-through in percentage changes). *Up to a first-order approximation, the percentage change in the price of firm f in industry i can be written as*

$$d \log p_{fi} = (1 - \alpha_{fi})(1 - \pi_{fi})d \log mc_{fi} + \pi_{fi} \frac{p_{-f,i}}{p_{fi}} d \log p_{-f,i}, \quad (35)$$

where

$$1 - \alpha_{fi} = \frac{mc_{fi}}{p_{fi}} \in [0, 1] \quad (36)$$

is the share of price due to marginal costs.

Proof. See Appendix B.2. □

Proposition 2 shows that when there is complete pass-through in levels, $\pi_{fi} = 0$, there is still incomplete pass-through in percentage changes since $1 - \alpha_{fi} \leq 1$. More generally, $\pi_{fi} \neq 1$. Thus, the presence of oligopolistic competition when firms face a logit-demand structure adds an additional mechanism through which changes in marginal costs can affect firm prices captured by the competitor's price channel $\pi_{fi} \frac{p_{-f,i}}{p_{fi}} d \log p_{-f,i}$.

We are now ready to examine how changes in import prices affect PPI. In our model, changes in import prices enter into two places: (i) through marginal cost, as an input in the production function, and (ii) as changes in the price of competitors in an industry (as it contains $d \log p_{Fi}$). The following proposition introduces a perturbation ($d \log mc_{fi}, d \log p_{Fi}$) and provides an expression for industry level PPI changes:

Proposition 3. *Changes in the producer price index after a perturbation ($d \log mc_{fi}, d \log p_{Fi}$)*

¹Note that to write $d \log mc_{fi} = d \log mc_i$ we require all firms to have the same initial marginal cost. This can still lead to different market and variety shares provided that there is heterogeneity in z_{fi} .

satisfy

$$\begin{aligned} d \log p_i^d &= \frac{1}{\left(1 - (1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})\right)\right)} \left(v_i^d \left(\sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi} \right) \right), \\ &+ \frac{1}{\left(1 - (1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})\right)\right)} \left(\frac{(1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})\right)}{(1 - s_{Fi})} s_{Fi} d \log p_{Fi} \right) \end{aligned} \quad (37)$$

where

$$\begin{aligned} \tilde{\delta}_{fi} &= \frac{mc_{fi} q_{fi}}{VC_i^d}; & v_i^d &= \frac{VC_i^d}{R_i^d}; & \rho_{fi} &= \frac{(1 - \pi_{fi})^2}{1 - \pi_{fi} + \pi_{fi}^2} \in [0, 1] \\ \tilde{\pi}_{fi} &= \frac{\pi_{fi}}{\sum_{h \in \mathcal{D}_i} \pi_{hi}}; & 1 - \pi_{Fi} &= \sum_{f \in \mathcal{D}_i} \pi_{fi} \end{aligned}$$

Proof. See Appendix B.3. □

There are some parameters that deserve discussion from the proposition. The parameter ν_i^d represents the share of domestic variable costs in domestic revenues; this is the industry level counterpart of $1 - \alpha_{fi}$. The parameter ρ_{fi} captures a “firm cost pass-through in levels” as it is an increasing function of $(1 - \pi_{fi})$, the level marginal cost firm’s pass-through we found in Proposition 1.

The term $\sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi}$ is an industry-level shifter in marginal cost changes of all firms appropriately weighted. This industry level component have two sources of firm-level heterogeneity (beyond the idiosyncratic cost shifters): (i) initial cost shares of each firm in total industry costs ($\tilde{\delta}_{fi}$) and (ii) the cost pass-through in levels parameters (ρ_{fi}). The industry-level marginal cost shifter thus places more weight on larger firms (in terms of costs) and firms with higher cost pass-through in levels. This will play an important role in our empirical application. The own-cost pass-through in percentage changes at the industry level is thus the numerator $\nu_i^d \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi}$.

The second row shows the effect of an increase in foreign competition, captured by a change in the foreign firm’s price, $d \log p_{Fi}$. This effect can be understood by looking at two elasticities. The first elasticity is how domestic prices react to a change in the industry-level price; this effect is captured by $\frac{(1 - \pi_{Fi})}{1 - s_{Fi}} \sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})$ that depends on variety shares π_{fi} , foreign sales shares s_{Fi} , and cost pass-through ρ_{fi} . Of course, the pass-through of the industry price index on the producer price index will be larger whenever the competitors’ effect is stronger (higher $1 - \rho_{fi}$), (ii) sales to foreigners are larger (higher s_{Fi}) and (iii) varieties

assigned to domestic firms are an important fraction $(1 - \pi_{Fi})$. The second elasticity is the elasticity of the industry price index with respect to the foreign firm's price. This elasticity is simply $s_{Fi} d \log p_{Fi}$, the first-order effect of the foreign firm price on the aggregate industry price index.

The total effect of both perturbations also needs to account for the fact that changes in the producer price index affect the industry-level price index, which in turn affects the producer price index, and so on. The first term multiplying both perturbations can be written as

$$\frac{1}{1 - (1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi} (1 - \rho_{fi}) \right)} = \sum_{t=0}^{\infty} \left((1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi} (1 - \rho_{fi}) \right) \right)^t$$

and thus captures all these higher-order rounds.

We conclude this subsection by noting that Proposition 3 also delivers a corollary for the change in prices of expenditure categories. The sole difference between the models is that $\pi_{Fe} = 0$ for all e , as there is no foreign competition in retailers. The following corollary captures this:

Corollary 1. *Changes in the price index of category e , $d \log p_e$, after a change in firm level marginal costs ($d \log mc_{fe}$) satisfy*

$$d \log p_e = \frac{1}{\left(1 - \sum_{f \in e} \pi_{fe} (1 - \rho_{fe}) \right)} v_e \left(\sum_{f \in e} \delta_{fe} \rho_{fe} d \log mc_{fe} \right) \quad (38)$$

where

$$\delta_{fe} = \frac{mc_{fe} c_{fe}}{VC_e}; \quad v_e = \frac{VC_e}{R_e}; \quad \rho_{fe} = \frac{(1 - \pi_{fe})^2}{1 - \pi_{fe} + \pi_{fe}^2} \in [0, 1]$$

2.5.2 The Role of Industry Level Input-Output Networks

The preceding section provided a perturbation on the space $(d \log mc_{fi}, d \log p_{Fi})$. In our model, firms' marginal costs depends of firms' prices in other industries as well. We now provide an extension where we perturb intermediate import prices $d \log p_i^*$ and the price of foreign firms $d \log p_{Fi}$ and study how this propagates through the producer price index.

To characterize the producer price index response we make the following aggregation assumption:

Assumption 2 (Aggregation of input expenditure shares). *For every pair of industries i, j ,*

$$\begin{aligned} \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j} \omega_{fi,gj} &= \rho_i^d \Omega_{i,j} s_{gj}^d \quad \text{for every } g \in D_j, \\ \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j}^* &= \rho_i^d \Omega_{i,j}^*, \end{aligned}$$

where

$$\rho_i^d \equiv \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \quad \text{and} \quad \omega_{fi,gj} = \frac{p_{gj} x_{fi,gj}}{p_{fi,j}^x x_{fi,j}}$$

With Assumption 2 at hand, we can derive the following result (details are in Appendix A.1):

Proposition 4 (Producer price indices and input-output networks). *Consider a perturbation of imported input prices ($d \log \mathbf{p}$) and foreign competitors prices ($d \log \mathbf{p}_F$) keeping constant wages $d \log w = 0$. The response of producer price indices is such that*

$$d \log \mathbf{p}^d = \underbrace{\mathbf{\Phi}^* d \log \mathbf{p}^*}_{\text{Imported-Input Cost Channel}} + \underbrace{\mathbf{\Phi}^F d \log \mathbf{p}_F}_{\text{Foreign Competition Channel}}, \quad (39)$$

where

$$\mathbf{\Phi}^* = \mathbf{G} \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^* \quad (40)$$

$$\mathbf{\Phi}^F = \mathbf{G} (\mathbf{I} - \mathbf{S}_F)^{-1} (\mathbf{I} - \mathbf{\Pi}_F) (\mathbf{I} - \boldsymbol{\rho}^d) \mathbf{S}_F \quad (41)$$

$$\mathbf{G} = (\mathbf{I} - \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^d - (\mathbf{I} - \mathbf{\Pi}_F) (\mathbf{I} - \boldsymbol{\rho}^d))^{-1} \quad (42)$$

and

$$\mathbf{\Upsilon}^d = \text{diag}(\mathbf{v}^d); \quad \boldsymbol{\rho}^d = \text{diag}(\{\rho_i^d\}_{i \in \mathcal{I}}); \quad \mathbf{S}_F = \text{diag}(\mathbf{s}_F); \quad \mathbf{\Pi}_F = \text{diag}(\boldsymbol{\pi}_F),$$

are diagonal matrices and

$$\rho_i^d = \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi},$$

is a cost-weighted average of the cost pass-through parameters.

Proof. See Appendix B.4. □

Proposition 4 provides an industry level counterpart to Proposition 3. While Proposition

3 required detailed firm-level information, Proposition 4 only requires industry level data. Moreover, Proposition 4 decomposes changes in industry level producer price indices into an imported-input cost channel and a foreign-competition channel, while the earlier proposition was agnostic about the particular sources of variation on marginal costs.

Although it is heavy on notation, Proposition 4 provides a natural extension to our earlier results. There are two key differences. First, note that the higher order terms now take a matrix form as now producer price indices are linked across sectors via marginal costs. This production network structure is captured in $\Upsilon^d \rho^d \Omega^d$ in the propagation matrix $\mathbf{G}$, where Ω^d captures the exposure of industries to producer price indices in all other industries. The remaining objects in $\mathbf{G}$ are just the natural extension to matrices of the higher order rounds in Proposition 3. Apart from this modification, the first-round effect of foreign competition on producer price indices is identical to the one Proposition 4.

Second, Proposition 4 directly links the effect of imported intermediate input prices on producer price indices, which is captured by the matrix Φ^* . This matrix has a first-round component and a propagation component ($\mathbf{G}$). As we already explained the propagation component, we focus on the first-round effect. This first-round effect combines two components: (i) the own marginal cost exposure to imported intermediate inputs Ω^* , (ii) the cost pass-through in percentage changes matrix $\Upsilon^d \rho^d$. Intuitively, a rise in imported intermediate input prices raises industry's marginal cost by Ω_i^* (the i th row of the $\mathcal{I} \times \mathcal{I}$ matrix Ω^*). This marginal cost change then translates into a change in the producer price index of the industry of $v_i^d \rho_i^d$. This change in the producer price index of industry i then affect both producer price indices in all the other industries as well as the industry price level (inclusive of foreign firms), which in turn affect industries' marginal costs and so on. The matrix Φ^* captures these transmission mechanisms.

3 From Model Objects to Data

This section maps each model object to official US public data. The mapping is a contribution in its own right: it bridges customs records, input–output tables and price series that use different classification systems with no published concordance linking the entire chain. Two inputs have to be in hand before the price response of Proposition 4 can be evaluated: the border shock and the mapping into prices. Section 3.1 builds the input–output and bridge objects that turn a border shock into a predicted price change. Section 3.2 builds the border shock itself. Section 3.3 verifies that an observed tariff change can stand in for an unobserved border price change.

3.1 Mapping Economic Objects to PPI and PCE Data

We map the model to data using the Bureau of Economic Analysis (BEA) Input–Output accounts, which record intermediate input flows, import penetration, and final demand at the commodity level. Because the 2025 tariffs were country-specific, we first disaggregate imports by source region, combining Census trade composition at the product level with the BEA Trade in Value Added (TiVA) tables, which report region-level import totals by end use.²

Proposition 4 requires four matrices that we construct from the BEA Use, Make, and Import tables. The domestic and imported input coefficient matrices Ω^d and Ω^* govern, together with the Leontief inverse, how cost shocks propagate round by round through the production network. They are built by subtracting the Import matrix from the Use table to isolate domestic flows (and retaining it for foreign flows), normalizing by gross output, and converting from industry to commodity space via the Make table. The markup-adjustment matrix Υ^d is the ratio of variable cost to revenue for each commodity and scales how much of a cost shock passes through to prices.³ The foreign-competitor share $\mathbf{S}_F$ is each commodity’s import penetration ratio—imports divided by total domestic supply—and measures the initial market presence of foreign varieties. The remaining parameters of Proposition 4—the cost pass-through ρ^d and the foreign variety share Π_F —are not observed in the accounts. We pin Π_F with the following restriction:

Assumption 3. $\pi_{fi} = s_{fi}, \quad \forall f, i,$

so that a firm’s variety share equals its observed sales share and $\Pi_F = \mathbf{S}_F$; we describe our estimation of ρ^d below in Section 4.3. To carry predicted producer-price responses to consumer prices, we use the BEA’s PCE bridge, which maps each commodity into potentially several NIPA expenditure categories, as we use the TiVA import shares to split each category’s cost into a domestically sourced and an imported component. Category-specific wholesale, retail, and transportation margins then dilute the border shock as it reaches consumer prices.⁴

Linking the tariff schedule to these objects requires a three-step concordance: we use the HS-10-to-NAICS correspondence in the customs records to aggregate border price changes to the NAICS-6 level, then use the NAICS-to-BEA concordance and the Make matrix to assign each NAICS-by-country cell to a BEA commodity, and finally match BEA commodities to PPI series at the NAICS level. This chain maps approximately 95 percent of import value

²Appendix D.4 describes the region disaggregation procedure, which uses iterative proportional fitting to reconcile the two sources, and Appendix D.2 describes how we extend the 2017 Detail Benchmark tables to 2024.

³Under an alternative log-pass-through convention, $\Upsilon^d = \mathbf{I}$ and percentage marginal-cost changes pass through one for one. Appendix E.2 reports all results under this alternative.

⁴The margin dilution differs depending on whether pass-through is measured in logs or in levels; Appendix E.2 reports consumer-price results under both conventions.

onto a BEA commodity.⁵ Of the 402 Detail commodities, 273 can be matched to an observed PPI series—252 to a unique series and 21 to an average of multiple series. Commodities that lack a PPI series are not discarded: they remain in the input–output network and transmit tariff-induced cost changes to commodities that do have observed prices. Commodities with zero imports enter with zero predicted price change and serve as the comparison group in the regressions that follow.

3.2 The Border Shock and the Instrument

With the input–output and bridge mappings in hand, the remaining object is the border price shock $d \log p^*$. The model treats the imported-input cost channel and the foreign-competition channel as generated by different primitives. To operationalize both from a single observed series we impose the following restriction:

Assumption 4. $d \log p_i^* = d \log p_{Fi} = d \log(1 + \tau_i)$, $\forall i$, where $(1 + \tau_i)$ represents the *ad-valorem tariff applied to industry i* .

This assumption embeds two restrictions. First, it requires that tariffs pass through fully into duty-inclusive border prices, such that foreign exporters do not absorb part of the duty by cutting their pre-tariff prices. Section 3.3 tests this directly. Second, it equates the change in the imported input price in response to an industry-level tariff to the change in the foreign producer price the industry competes against. While the model allows for segmentation of these two markets, we lack data to distinguish them.

We construct the realized tariff shock from duties actually paid at the border (the *effective rate*), and we use the *statutory* schedule as an instrument. Both of these schedules are available from official Foreign Trade Census data. For every HS-10 product-level code p from country of origin c in month m , we compute the effective tariff rate by dividing the total duties paid by the total value of imports, and we compute the statutory observed tariff rate by dividing the total duties paid by dutiable values:⁶

$$\tau_{cp,m}^{\text{eff}} = \frac{\text{Duties Paid}_{cp,m}}{\text{Import Value}_{cp,m}} \quad \tau_{cp,m}^{\text{stat}} = \frac{\text{Duties Paid}_{cp,m}}{\text{Dutiable Value}_{cp,m}} \quad (43)$$

The effective rate is therefore the tariff schedule diluted by duty-free, exempt trade and by incomplete clearance; it is the rate importers actually paid. The statutory rate is the

⁵The unmatched 5 percent consists overwhelmingly of special-provision and low-value-shipment lines carrying no NAICS stamp at entry. We exclude them rather than reallocate them; Appendix D.3 reports coverage by base year.

⁶When a rate is not observed in a certain month, we carry forward the last observed rate.

schedule as applied only to the share of import value that is exemption-free; it tracks the legal rate more closely but can still diverge from the enacted schedule when collection lags behind legislation. The gap between the two is the main identification concern: exemption qualifications can shift endogenously with firms’ import decisions.

We compare these two rates to a third tariff schedule: the *enacted* rate. This rate is sourced from the official Federal Register and USITC and records official law changes at their effective date of implementation, rather than at the first month in which a statutory rate is observed. Figure 1 shows the comparison of the three series since January 2025. Each series is aggregated on fixed 2024 import-value shares so that the series are not affected by substitution away from heavily tariffed product-country combinations. Statutory rates track the enacted schedule closely except in April and May 2025. At this time, a 125% tariff on China was officially held in place for 34 days and then paused retroactively. Because it was paused before most ships landed, few importers actually paid it. The gap between statutory and effective rates is much larger.

We use the statutory tariff rate as the main instrument rather than the enacted rate for two reasons. First, the local projection requires a well-measured monthly shock, and a large but largely unpaid tariff distorts estimates. Second, the gap between the enacted and statutory series is concentrated on country-product combinations that trade infrequently and are more likely to be unobserved in any given month, which generates timing discrepancies between the legislative effective date and the first observed rate at the border. Because our interest is in tariff effects at a high level of industry aggregation, most of the identifying variation comes from country-products whose statutory rates are always observable.⁷

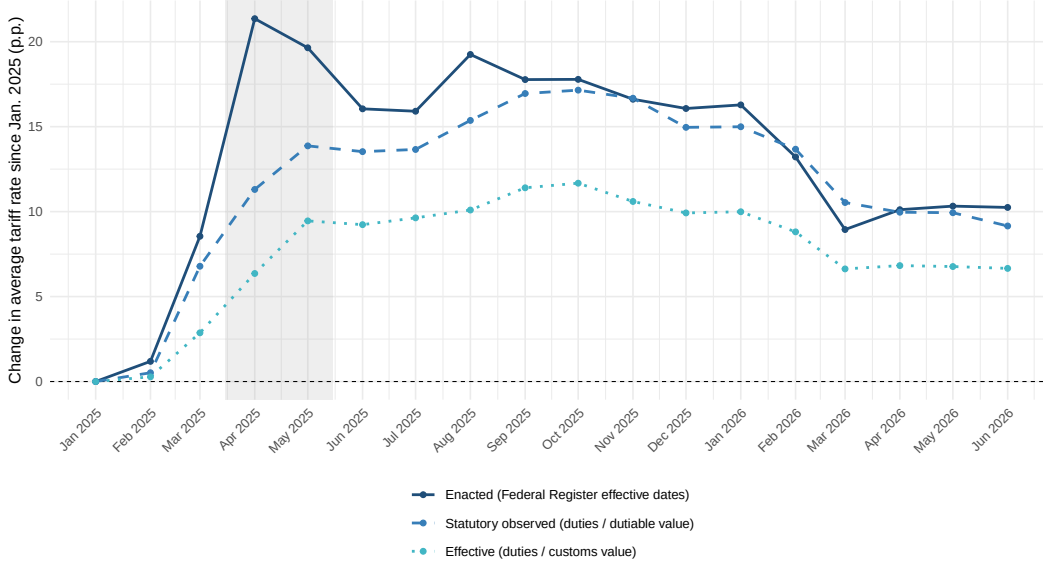
3.3 Border Pass-Through

We test full tariff pass-through into border prices at the individual import record level. We define a record k as a unique HS-10 product-by-country-of-origin combination. Let v_{kt} denote the duty-inclusive unit value of record k , and let $\Delta_h \log v_{k,t+h} \equiv \log v_{k,t+h} - \log v_{k,t-1}$. For each horizon h we run a local projection in differences, following Dube et al. (2025):

$$\Delta_h \log v_{k,t+h} = \alpha_k + \delta_{c(k),t} + \beta_h \Delta \log(1 + \hat{\tau}_{k,t}^{\text{eff}}) + \sum_{\substack{m=-(h-1) \\ m \neq 0}}^{12} \kappa_{h,m} \Delta \log(1 + \tau_{k,t-m}^{\text{stat}}) + \varepsilon_{k,t+h}, \quad (44)$$

⁷Appendix E.5 re-estimates pass-through using the enacted schedule rate as the instrument and describes how the April–May episode is handled. Since the two series agree on the size of the tariff and disagree primarily on when it happens, that comparison is run on a cumulated tariff over a window. On that basis, the two instruments agree (Table E.1).

Figure 1: Monthly Tariff Shock: Enacted, Statutory Observed and Effective



Notes: Change since January 2025 in the average tariff rate, aggregated across HS-10 product $\times$ country-of-origin pairs on import weights fixed at their 2024 values. The three lines' measurement is described in the main text. Shading spans mid-March to mid-May 2025, the window of the China increase that was imposed and then paused retroactively.

where the effective tariff change at t is instrumented by its statutory counterpart. We include product-by-origin fixed effects α_k as well as origin-by-month fixed effects $\delta_{c(k),t}$, which absorb the bilateral exchange rate and any other country-level response. Lags and leads of the statutory path control for the serially correlated nature of the 2025 tariff waves. Standard errors are two-way clustered by HS-8 product and by country of origin.

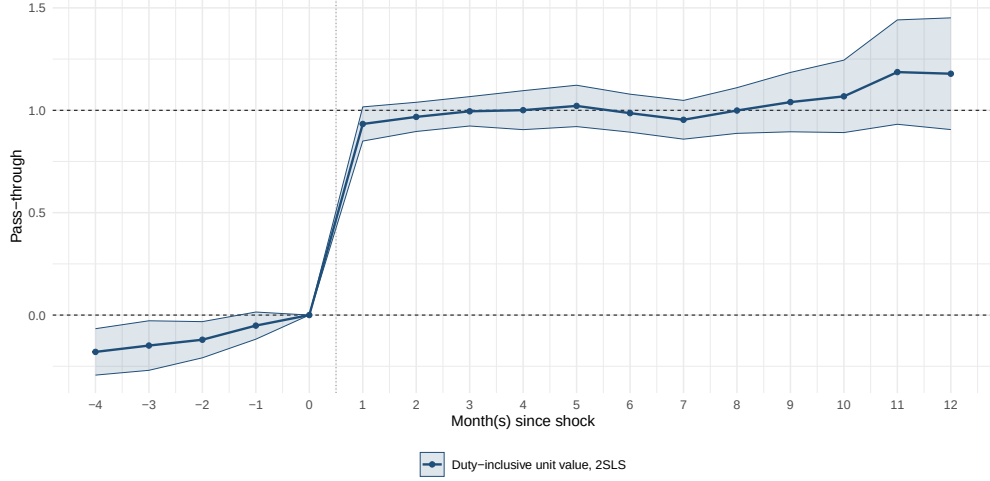
Figure 2 plots $\hat{\beta}_h$. Pass-through is complete on impact and stays complete: 0.93 in the month after the shock and 0.99 by the third month, inside a narrow band around one for the rest of the year.⁸

Because tariffs pass through fully at the border, we can treat the observed tariff change $d \log(1 + \tau)$ as the empirical counterpart of the model's foreign-price shock $d \log \mathbf{p}^*$.⁹ We next ask whether the model maps that shock into domestic producer prices.

⁸The estimation includes three million import records at the first horizon and 1.9 million at the twelfth. Restricting attention to records that trade continuously across the whole twelve months leaves the path unchanged (Appendix C).

⁹The tariff is also the sharper of the two available measures: declared unit values carry compositional and reporting noise that the tariff rate does not, so using the tariff instead of the unit value reduces measurement error in the shock and raises the power of our regressions. Appendix E.5 considers the duty-inclusive unit value as an alternative treatment for our main specification.

Figure 2: Border Pass-through of Tariffs into Import Unit Values, 2025



Notes: $\hat{\beta}_h$ from (44), estimated on monthly import records at the HS-10 product $\times$ country-of-origin level. The dependent variable is the log change in the duty-inclusive unit value from $t - 1$ to $t + h$. The dashed line at one marks complete pass-through of the duty into the border price. Month 0 is the reference period. Shading gives 95% confidence bands, two-way clustered by HS-8 product and country of origin. Sample is unbalanced so it narrows from 3.0 million records at $h = 1$ to 1.9 million at $h = 12$.

4 Producer-Price Evidence

We now turn to understanding how the model’s mapping of the border shock into domestic producer prices holds up in the data. To do so, we build three nested versions of the model’s predicted response by successively activating the channels of the model. At each step, we assess how well the additional channel does in explaining the observed changes in the data. This allows us to understand the importance of imported-input costs alone, network propagation, and foreign competition.

4.1 Model-Implied Producer-Price Responses

The object we take to the data is the model’s own producer-price response, evaluated on measured input–output matrices and the observed tariff change. Under Proposition 4 it is

$$\Delta \log \mathbf{p}^M(\boldsymbol{\rho}^d) = [\mathbf{I} - (\mathbf{I} - \boldsymbol{\rho}^d)(\mathbf{I} - \mathbf{S}_F) - \boldsymbol{\rho}^d \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^d]^{-1} [\boldsymbol{\rho}^d \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^* + (\mathbf{I} - \boldsymbol{\rho}^d) \mathbf{S}_F] \Delta \log(1 + \tau) \quad (45)$$

where the border shock is replaced by the observed tariff change under the assumption of Section 3.2. Two objects in (45) govern how much a border shock is amplified before it reaches a producer price: the domestic network $\boldsymbol{\Omega}^d$ and the conditional own-cost pass-through parameter $\boldsymbol{\rho}^d$. Our empirical strategy compares observed price changes with three nested

versions of $\Delta \log \mathbf{p}^M$, obtained by successively activating those two objects.¹⁰

First-round imported-input cost response. Setting $\boldsymbol{\rho}^d = \mathbf{I}$ and setting the off-diagonal elements of $\boldsymbol{\Omega}^d$ to zero gives

$$\Delta \log \mathbf{p}^{M,\text{fr}} = \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^* \Delta \log(1 + \tau). \quad (46)$$

Each industry responds only to the import prices it directly purchases, with no round-by-round propagation through other industries. This object is most comparable to a naive import-share calculation of tariff exposure.

Network-adjusted cost response. Here, we consider the full matrix $\boldsymbol{\Omega}^d$ but still set $\boldsymbol{\rho}^d = \mathbf{I}$:

$$\Delta \log \mathbf{p}^{M,\text{net}} = (\mathbf{I} - \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^d)^{-1} \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^* \Delta \log(1 + \tau). \quad (47)$$

The inverse matrix $(\mathbf{I} - \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^d)^{-1}$ captures the propagation of these cost changes through the domestic production network. The gap between equation (46) and equation (47) quantifies network propagation: the extent to which an import price shock is magnified as it passes through successive rounds of domestic intermediate-input purchases beyond each industry's first-round imported-input exposure.

Network-adjusted costs plus foreign competition. Imposing a common conditional own-cost pass-through $\boldsymbol{\rho}^d = \bar{\rho} \mathbf{I}$ in (45) gives

$$\Delta \log \mathbf{p}^M(\bar{\rho}) = [\mathbf{I} - (1 - \bar{\rho})(\mathbf{I} - \mathbf{S}_F) - \bar{\rho} \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^d]^{-1} [\bar{\rho} \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^* + (1 - \bar{\rho}) \mathbf{S}_F] \Delta \log(1 + \tau). \quad (48)$$

As $\bar{\rho}$ falls below one, the foreign-competition channel through $\mathbf{S}_F$ becomes active and domestic prices respond directly to border-price movements in addition to the imported-input cost channel through $\boldsymbol{\Omega}^*$.¹¹ At the industry level, each producer price becomes a weighted average of its own marginal-cost response and the foreign-competitor price change:

$$d \log p_i^d = v_i^d \tilde{\rho}_i^d \mathbb{E}_{\tilde{\delta}_i} [d \log \mathbf{mc}_i] + (1 - \tilde{\rho}_i^d) d \log p_{Fi}, \quad (49)$$

¹⁰The headline empirics report pass-through in levels, following the model. However, pass-through in logs ($\boldsymbol{\Upsilon}^d = \mathbf{I}$) is carried through every exercise and reported in Appendix E.2, and Section 4.5 explains why the two bracket rather than compete.

¹¹Unlike the first two measures, this measure has a free parameter. In Section 4.3, we describe how we estimate $\bar{\rho}$.

where $\tilde{\rho}_i^d = \rho_i^d / (\rho_i^d + (1 - \rho_i^d)s_{Fi})$. How much domestic prices move relative to marginal cost depends on the sales share of foreign competitors within i and the tariff hike in i .¹² The conditional own-cost pass-through parameter ρ_i^d depends on market structure and the elasticity of substitution between products.

Because $\bar{\rho}$ enters (48) inside the network inverse, it reshapes which commodities the network treats as exposed rather than scaling the predicted response by a common factor. The foreign-competition channel and network propagation therefore interact rather than adding up, which is the property the estimator in Section 4.3 exploits.

4.2 Local-Projection Pass-Through

We test how each of these predicted responses compares to the actual data using local projections following Jordà (2005) in the difference-in-differences design of Dube et al. (2025). For each horizon $h \in \{1, \dots, H\}$ with $H = 12$ we estimate

$$\begin{aligned} \Delta_h \log p_{i,t+h} = & \alpha_i + \delta_t + \beta_h \Delta \log p_{i,t}^M + \sum_{\substack{k=-(h-1) \\ k \neq 0}}^H \kappa_{h,k} \Delta \log p_{i,t-k}^{M,\text{stat}} + \lambda_h \Delta \log w_{i,t-1} \\ & + \sum_{k=0}^H \psi_{h,k} o_{i,t-k} + \varepsilon_{i,t+h}, \end{aligned} \quad (50)$$

where $\Delta_h \log p_{i,t+h} \equiv \log p_{i,t+h} - \log p_{i,t-1}$ is the cumulated log-price change from $t-1$ to $t+h$, $\Delta \log p_{i,t}^M$ is the model-implied price response of item (or industry) i from Section 4.1, $\Delta \log p_{i,t}^{M,\text{stat}}$ is its statutory counterpart, and $\Delta \log w_{i,t-1}$ is one lag of wage growth. Each predicted response is instrumented with its counterpart constructed from statutory rates. The leads and lags account for the serially correlated nature of the 2025 tariff waves, while item and date fixed effects isolate the cross-sectional variation in tariff exposure, absorbing general equilibrium forces and category-specific pre-trends. In this section, we estimate the projection on the PPI panel, where i is a NAICS commodity; in the next section, we estimate it on the PCE panel, where i is a NIPA expenditure category. We consider shocks between 2024 and August 2025, the last month for which the effect can be projected over the following 12 months, and we hold the sample fixed across all 12 horizons.¹³ Standard errors are clustered by item.

¹²Appendix A.2 analyzes the role of the within-industry covariance between firm-level pass-through and import exposure. It provides conditions under which this covariance reduces to a single industry-level parameter γ_i . In the strategy of this section, however, these parameters are absorbed by our estimated $\bar{\rho}$. Our generalized methods of moments (GMM) moment conditions exploit cross-sectional variation in import exposure that would otherwise be attributed to γ , rendering a separate parameter unnecessary.

¹³This gives us 5,145 observations and 258 items.

The regressor is a price change predicted by the model, not a tariff rate, so β_h is a test of the prediction rather than a pass-through elasticity: $\beta_h = 1$ says that by horizon h observed prices have moved by the amount the model predicts, $\beta_h > 1$ says the model understates the pressure, and $\beta_h < 1$ says pass-through is incomplete through the lens of the model.

4.3 Estimating Strategic Complementarities

As discussed, the first two predicted responses contain no free parameters: they follow from the input–output tables and the observed tariff change. However, the third introduces $\bar{\rho}$, the average conditional own-cost pass-through, which we do not observe. We estimate it by generalized method of moments (GMM), requiring that the long term residuals from equation (50) satisfy two model-implied conditions.

Let $\tilde{\cdot}$ denote residuals after projecting on the fixed effects and controls of equation (50), and let $y_{i,t+h_{\text{ref}}}$ be the h_{ref} -month forward log price change. The residual at the reference horizon implied by the model under $\bar{\rho}$ is $u_{i,t}(\bar{\rho}) = \tilde{y}_{i,t+h_{\text{ref}}} - \widetilde{\Delta \log p_{i,t}}^M(\bar{\rho})$. We impose two moment conditions on it:

$$\underbrace{\mathbb{E} \left[\widetilde{\Delta \log p_{i,t}}^{M,\text{stat}}(\bar{\rho}) u_{i,t}(\bar{\rho}) \right]}_{\text{correct scale}} = 0, \quad \underbrace{\mathbb{E} \left[\left(\frac{\partial \widetilde{\Delta \log p_{i,t}}^{M,\text{stat}}}{\partial \bar{\rho}} \right) u_{i,t}(\bar{\rho}) \right]}_{\text{correct tilt}} = 0. \quad (51)$$

The first moment requires that the model predict the right overall *magnitude* of price pressure: a unit of predicted price pressure should move producer prices one-for-one. The second moment requires that the model predict the right cross-sectional *shape*: lowering $\bar{\rho}$ raises predicted pressure more for commodities exposed to foreign competition than for those exposed only through input costs, and the second moment asks that this reallocation match the data. The parameter is thus identified from the model’s implied cross-sectional pattern of tariff pressure, not chosen to force the aggregate pass-through to one.

The identification assumption this requires is a conditional-mean restriction, $\mathbb{E}[u_{i,t}(\bar{\rho}_0) | \mathcal{Z}] = 0$, where $\mathcal{Z}$ is the exogenous statutory information set. In other words, no force outside equation (45) can move prices in a way that lines up either with the statutory tariff variation or with the way the network redistributes it across commodities.¹⁴

This design has two advantages over estimating $\bar{\rho}$ directly from the interaction of sales shares and tariff shocks, $\mathbf{S}_F d \log(1 + \boldsymbol{\tau})$, as in [Amiti, Redding and Weinstein \(2019\)](#) and [Flaen and Pierce \(2024\)](#). First, the network nature of our model implies that strategic complementarities interact with marginal costs and the structure of the economy nonlinearly;

¹⁴Appendix F relaxes these restrictions and shows that the estimated $\bar{\rho}$ is stable.

neglecting network spillovers is a SUTVA violation. Second, we do not allow the sensitivity of prices to sales shares to be completely decoupled from the sensitivity to marginal cost. While our model allows—to some extent—for nonhomogeneity of the two sensitivities via pass-through in levels, we still impose the restriction that a high sensitivity to $\mathbf{S}_F$ implies a lower sensitivity to marginal cost via $\mathbf{\Omega}^*$.

Table 1 presents the GMM estimate and inference statistics. The two conditions pin down $\hat{\rho} = 0.655$ (95% confidence set $[0.433, 0.886]$). We directly test what happens when we shut down the foreign competition channel by setting $\bar{\rho} = 1$; this benchmark is rejected at the 1% level ($S = 10.35$, $p = 0.006$). This shows how without strategic complementarities, the model cannot match both the overall magnitude and the cross-sectional pattern of price changes.

Table 1: GMM Estimate of the Common Conditional Own-Cost Pass-through $\bar{\rho}$

Quantity	Estimate
$\bar{\rho}$	0.655
95% confidence set	$[0.433, 0.886]$
β_9 at $\bar{\rho} = 1$	2.314 (0.537)
β_9 at $\hat{\rho}$	1.173 (0.271)
$J = S(\hat{\rho})$, p (1 df)	1.56 [0.211]
$S(\bar{\rho} = 1)$, p (2 df)	10.35 [0.006]
First-stage F at $\hat{\rho}$	10315.4
Observations	5,145
Items (clusters)	258

Notes: One common average marginal cost pass-through $\bar{\rho}$ estimated from the two moments of (51), at reference horizon 9. $\hat{\rho}$ minimises the item-clustered criterion (F.6) on a 0.02 grid over $[0.02, 1]$, and the 95% confidence set collects the values that criterion does not reject, $\{\bar{\rho} : S(\bar{\rho}) \leq \chi_{2,0.95}^2 = 5.99\}$. The rows β_9 at $\bar{\rho} = 1$ and at $\hat{\rho}$ are local-projection coefficients with item-clustered standard errors in parentheses. $J = S(\hat{\rho})$ tests $\beta_9 = 1$ against $\chi_{1,0.95}^2 = 3.84$, and $S(\bar{\rho} = 1)$ tests that restriction jointly with $\bar{\rho} = 1$ itself against $\chi_{2,0.95}^2$; p -values in brackets.

At $\hat{\rho} = 0.655$, a producer’s pricing decision places about two-thirds of the weight on its own marginal cost and the remaining third on the price of the imported varieties it competes with. Other papers have estimated the same weight directly from firm and competitor prices (Atkeson and Burstein, 2008; Amiti, Itskhoki and Konings, 2019). Overall, we find our estimate to be in line with the literature: Amiti, Itskhoki and Konings (2019) estimate that a typical firm places roughly 0.6 of the weight on its own costs, which lies within our confidence set, and 0.4 on competitors’ prices.

Note that, our estimate comes from industry-level data, while the pricing decision the parameter describes is a firm-level one. Appendix F works with the general version of the

model that makes the cost of that aggregation explicit. Averaging firm pricing rules within an industry leaves a term beyond $\bar{\rho}$: the within-industry covariance between a firm’s own-cost pass-through and its exposure to imported inputs, collected in $\mathbb{C}^*$. Estimation at the industry level that assumes $\mathbb{C}^* = 0$, as above, may suffer from omitted variable bias. Appendix F shows how, under relatively mild assumptions, we can estimate both $\bar{\rho}$ and $\mathbb{C}^*$. Doing so gives $\hat{\rho} = 0.719$ and again rejects $\bar{\rho} = 1$.¹⁵

We also estimate $\bar{\rho}$ using the parametric specification of (F.10), letting ρ vary with market structure as the microfoundation implies. Decreasing ρ in the import-adjusted Herfindahl returns a gradient of the predicted sign and magnitude and a median ρ_j of 0.652, close to the common- ρ estimate (Appendix F.4). These industry-level moments have too little power, however, for the parameterization to count as independent corroboration.

4.4 Pass-Through Results

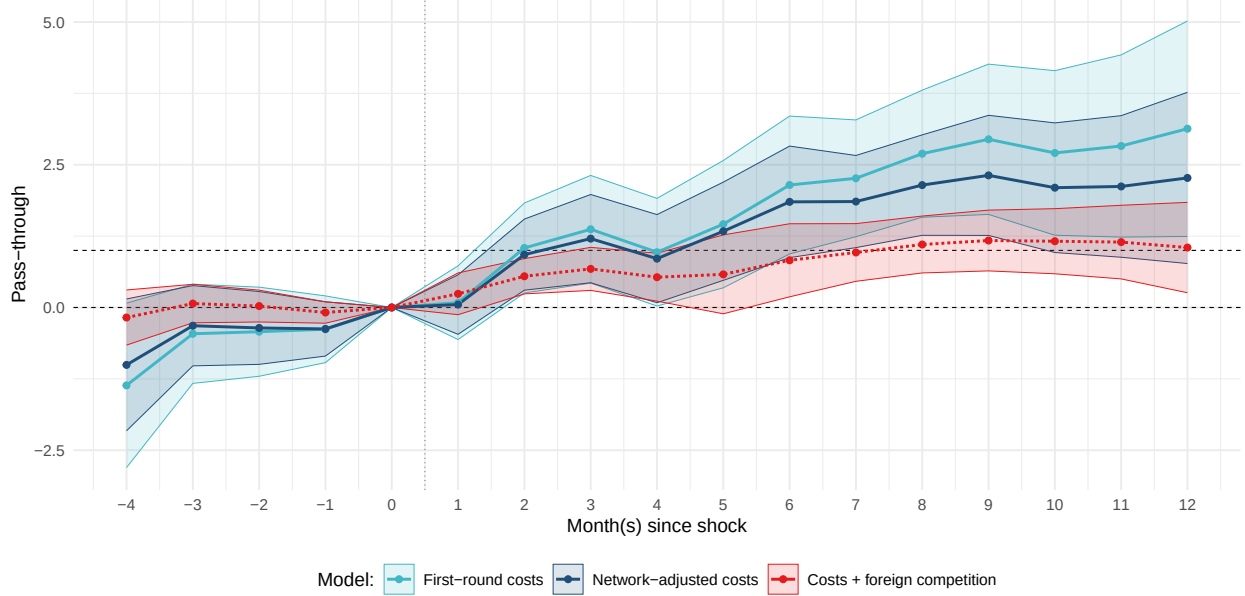
Figure 3 summarizes the nested local projection exercise. We evaluate each model of predicted price response in turn, taking horizon 9 as the reference for the long-term pass-through. For the first-round cost response we get $\hat{\beta}_9 = 2.946$ (0.672), and once domestic network propagation is added we get $\hat{\beta}_9 = 2.314$ (0.537). Both are rejected against $\beta_9 = 1$ ($p = 0.004$ and $p = 0.014$, respectively). The finding that both coefficients exceed one implies that neither of these models can account for full price movements. Network propagation raises the model-implied response, but the parameter-free models still predict substantially less producer-price pressure than appears in the data. Switching on the foreign-competition channel at $\hat{\rho}$ brings the producer-price coefficient to $\beta_9 = 1.173$ (0.271), which is within one standard error of one; the restriction $\beta_9 = 1$ is not rejected (see Table 1). Of the distance between the first-round response and one-for-one pass-through, network propagation covers about a third and the foreign-competition channel a further three-fifths.

The movement of domestic prices by more than own-cost exposure implies is also a feature of the 2018–19 evidence. Amiti, Redding and Weinstein (2019) and Fajgelbaum et al. (2020) find that the domestic prices of import-competing goods rose alongside the tariffed import prices, and Flaaen and Pierce (2024) document the same protective effect in industry-level US manufacturing producer prices. Amiti, Heise and Weinstein (2026) report it for the 2025 episode in matched import and producer prices. Our β_h multiplies a model-implied price change rather than a tariff rate, so the coefficients are not directly comparable, but the direction is the same. In our setting, prices move more than the model predicts even after

¹⁵The covariance term is imprecisely estimated. Studies of pass-through heterogeneity typically find that the firms most exposed to imported inputs pass through *less* than their industry average, so the covariance is usually negative. A negative covariance delivers a $\bar{\rho}$ below 0.65 rather than overturning it.

accounting for network propagation.

Figure 3: PPI Tariff Pass-through — First-Round Costs, Network-Adjusted Costs, and Foreign Competition, 2025 Episode



Notes: $\hat{\beta}_h$ from (50) for the 2025 episode, producer prices, pass-through in levels; item i is a BEA industry. Each line uses a different model-implied exposure from (49) and is estimated in its own regression: the first-round imported-input cost response and the network-adjusted cost response, both at $\rho = 1$, and the network-adjusted response with the foreign-competition channel evaluated under $\rho = \hat{\rho}$. Dashed lines mark zero and complete pass-through. Bands are 95% confidence intervals clustered by item.

Switching on the foreign-competition channel also improves fit and pre-trend behavior. First, fit: the partial R^2 goes from 0.0023 for the first-round response, 0.0022 with network propagation added and 0.0033 at $\hat{\rho}$.¹⁶ Second, pre-trend: under the network-adjusted cost measure there is some evidence that producer prices were already moving against the schedule before it took effect; rebuilding the predicted response at $\hat{\rho}$ flattens the pre-period. The industries the foreign-competition-augmented model predicts to move most are not the ones already drifting upward before the tariffs were introduced.

4.5 Validation and Interpretation

Finally, we run the same exercise on the 2018–19 waves. That episode presents more identification issues than 2025: the specification fails the pre-trend test at every horizon

¹⁶The partial R^2 values are small in magnitude because item and date fixed effects and the control block absorb almost all of the variance of monthly producer-price changes; only the comparison across the three measures is informative.

and standard errors are roughly two and a half times as large. Both the network and the competition channel still matter, and pass-through is again centered on one: the first-round measure gives 2.867 (1.725), network propagation 2.232 (1.164), and the foreign-competition channel $\beta_9 = 1.248$ (0.679). The criterion is minimized at $\hat{\rho} = 0.697$, close to the 2025 estimate, but the confidence set $[0.238, 1.000]$ is wide enough that the episode neither corroborates nor contradicts it—in particular, $\bar{\rho} = 1$ is *not* rejected here ($S = 1.38$ against $\chi^2_{2,0.95} = 5.99$), so the price-taking benchmark survives on this sample. We therefore do not use 2018–19 to discipline ρ ; Appendix E.1 reports it in full.

Appendix E examines the remaining sensitivity of the local-projection results along one dimension at a time: the sign-symmetry of negative and positive tariff shocks, the sample of consumption categories, and the choice of instrument. The estimates are not identical across specifications, but they support the same qualitative interpretation as the baseline.

4.6 Why Strategic Complementarities Amplify the 2025 Tariff Shock

Including strategic complementarities lowers the estimated pass-through coefficient by raising the model-implied response to tariffs. This may sound counterintuitive because strategic complementarity is usually the force that attenuates shocks to marginal costs: a firm that cares about its rivals’ prices absorbs part of a cost increase in its margin instead of passing it on (Atkeson and Burstein, 2008). In this model, however, tariffs directly affect both marginal costs and competitors’ prices. Whether including strategic complementarities raises or lowers the model-implied response relative to the price-taking benchmark depends on the changes in the full tariff schedule, and it is episode specific.

Rewrite the industry-level pass-through equation (49) so that the two forces at work appear separately,

$$d \log p_i^d = v_i^d \mathbb{E}_{\delta_i}[d \log \mathbf{mc}_i] + (1 - \tilde{\rho}_i)[d \log p_{Fi} - v_i^d \mathbb{E}_{\delta_i}[d \log \mathbf{mc}_i]] . \quad (52)$$

The first term is the own-cost response. The second is the foreign-competition channel, and its weight $1 - \tilde{\rho}_i$ rises as ρ falls. Its *sign* is the sign of the bracket, which compares the price change of the imported varieties industry i competes against with the change in i ’s own marginal cost.

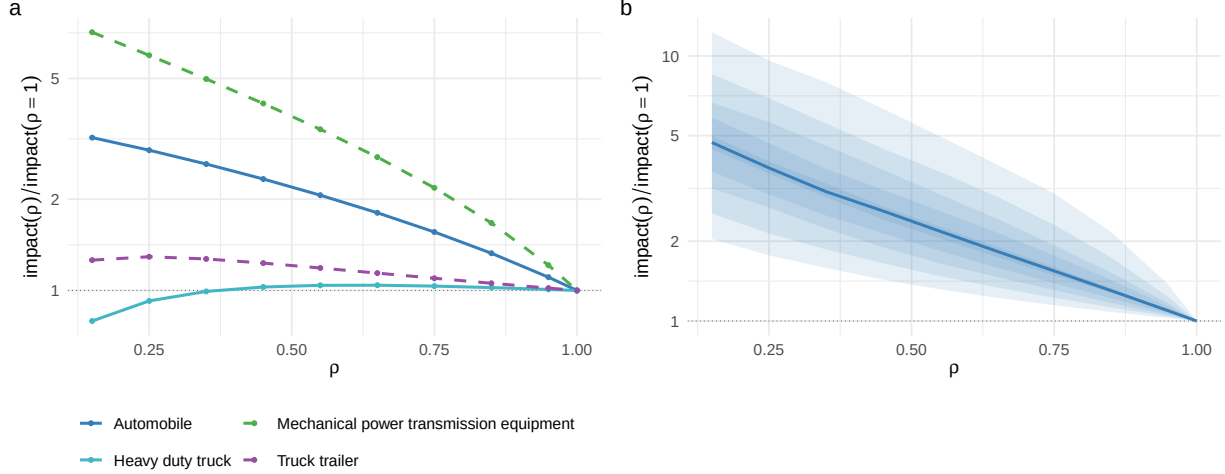
The familiar case is a shock that reaches industry i through its costs only. Its inputs become more expensive while the imported goods it competes with do not, the bracket is negative, and lowering ρ lowers the predicted price response: the firm holds its price closer to its unchanged rivals’ and takes the cost increase out of its margin. Read through the network,

that is a shock to Ω^* with no accompanying shock to $\mathbf{S}_F$.

The 2025 schedule mostly did not have this marginal-cost-specific effect. Tariffs applied mostly at the country-level, with widespread effects felt in all industries, so for most commodities the imported varieties a domestic producer competes with became more expensive at the same time as its inputs did, and $d \log p_{Fi}$ is large and positive. The term in bracket then turns positive: the foreign-competition channel adds to the own-cost term instead of offsetting it, and lowering ρ *raises* the predicted impact. This is the protectionist effect that [Amiti, Redding and Weinstein \(2019\)](#) and [Flaaen and Pierce \(2024\)](#) document for the 2018–19 tariffs, here operating on top of network cost propagation rather than on its own.

Figure 4 shows how much the channel amplification varies widely across commodities. Panel (a) isolates the mechanism with a pair of commodities that differ in little else. Automobiles and heavy duty trucks are both NAICS 3361 assemblers drawing on the same steel- and aluminium-intensive input basket, so their imported-input cost channels are alike, but the April 2025 Section 232 action covered the finished automobile and carved heavy duty trucks out; over the estimation window the cumulated foreign-competitor price change is 13.4 percentage points for automobiles against 1.6 for trucks. With both markets shocked the automobile path rises as ρ falls, and with only the cost market shocked the truck path falls. Two further commodities, selected by rule from the same grid, give the range: at the lowest admissible gridpoint, $\rho = 0.15$, the predicted impact is 7.8 times its Leontief value for mechanical power transmission equipment and 1.2 times for truck trailers. Panel (b) pools all 232 manufacturing commodities. The channel amplifies for all but one of them: heavy duty trucks are the only commodity whose predicted impact is smaller at $\rho = 0.15$ than at $\rho = 1$.

Figure 4: ρ -sensitivity of the predicted PPI impact — 2025 Episode



Notes: Both panels plot the predicted PPI impact at ρ , divided by the same commodity's impact at $\rho = 1$. The vertical axis is logarithmic. Impacts are cumulated over January–October 2025 under pass-through in levels, on Detail BEA commodities with commodity-specific source-region import weights (Appendix D.4). Panel (a): automobiles (336111) and heavy duty trucks (336120), which share the NAICS 3361 input basket but differ in exposure to competing imports, together with mechanical power transmission equipment (333613) and truck trailers (336212), the steepest and the flattest path among commodities in the top quartile of [impact at $\rho = 1$]. Panel (b): all 232 manufacturing commodities, with the cross-commodity median and shaded ribbons at the 45–55, 35–65, 25–75, 15–85 and 5–95 percentile ranges.

5 Consumer-Price Evidence

5.1 Mapping Producer Prices into PCE Categories

The producer-price estimates identify the production block. We now carry the same model-implied producer response through the retail mapping to PCE categories. Let $\Theta^d = [\theta_{ej}^d]_{e \in \mathcal{E}, j \in \mathcal{I}}$ and $\Theta^* = [\theta_{ej}^*]_{e \in \mathcal{E}, j \in \mathcal{I}}$ denote expenditure-category sourcing matrices, where θ_{ej}^d (θ_{ej}^*) is the cost share of domestic (imported) commodity j in the marginal cost of expenditure category e . The retailer marginal-cost vector is then

$$d \log \mathbf{mc}^r = \Theta^d d \log \mathbf{p}^d + \Theta^* d \log \mathbf{p}^*. \quad (53)$$

The matrix Θ^d captures retailers' exposure to domestically produced commodities, whose prices already reflect imported-input shocks propagated through the domestic production network. The matrix Θ^* captures retailers' exposure to imported finished commodities. Because all firms within an expenditure category are domestic retailers, there is no foreign-competitor-price term at this layer.

Under Corollary 1, the expenditure-category price vector satisfies $d \log \mathbf{p} = \mathbf{Y}^r d \log \mathbf{mc}^r$,

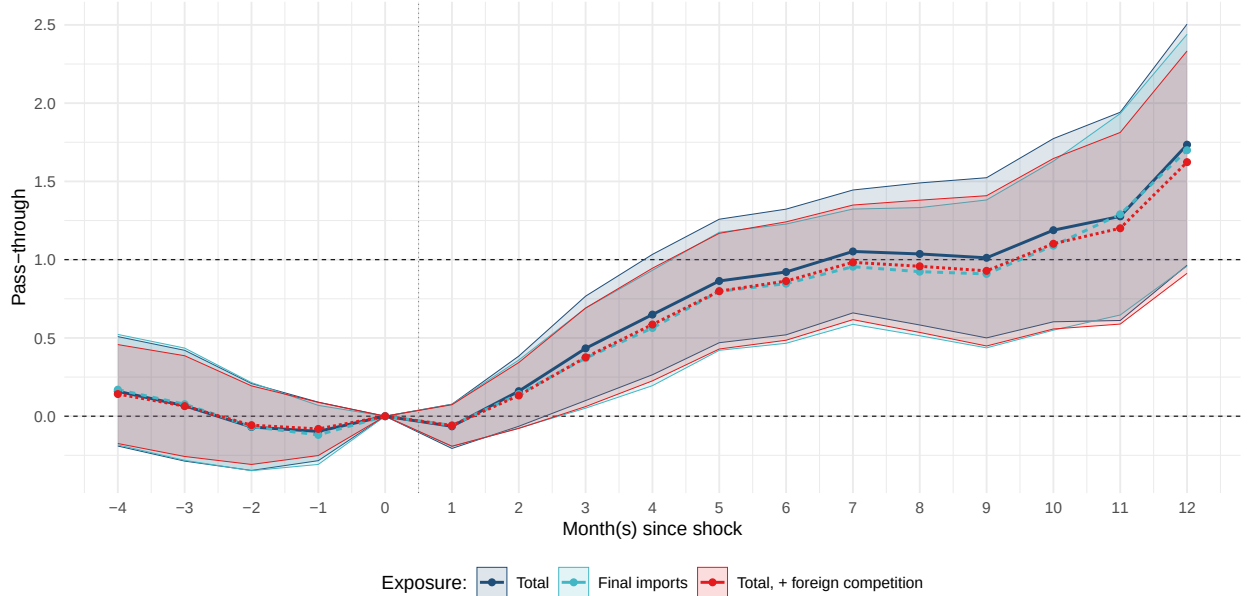
where Υ^r is the diagonal markup-adjustment matrix. Substituting the model-implied producer response (47) for $d \log \mathbf{p}^d$ yields

$$d \log \mathbf{p} = \Upsilon^r \left[\underbrace{\Theta^d (I - \Upsilon^d \Omega^d)^{-1} \Upsilon^d \Omega^*}_{\text{Domestic-Production Channel}} + \underbrace{\Theta^*}_{\text{Final-Import Channel}} \right] d \log \mathbf{p}^*. \quad (54)$$

Equation (54) decomposes the consumer-price response into two channels: a final-import channel, in which imported finished commodities enter retailer costs without passing through the domestic production network, and a domestic-production channel, in which border-price changes propagate through domestic producer prices and the input-output network into retailer costs. The key result of equation (54) is that all objects in the right-hand side are observable in the data. We can thus construct model implied price response $\Delta \log p_{e,t}^M$ of expenditure category e , which enters the projection (50) in place of the producer-side response.

5.2 PCE Pass-Through

Figure 5: PCE Tariff Pass-through — Total and Final-Import Exposure, 2025 Episode



Notes: $\hat{\beta}_h$ from (50) for the 2025 episode, PCE consumer prices, pass-through in levels; item i is a NIPA expenditure category. Total exposure and its final-import component are estimated in separate regressions and overlaid; the third line rebuilds the domestic-production component of total exposure at the $\hat{\rho}$ of Section 4.3. Estimator and controls as in Section 4.2, with the contemporaneous exposure term only. Dashed lines mark zero and complete pass-through. Bands are 95% confidence intervals clustered by item.

The PCE estimates in Figure 5 show that the model captures an important part of the cross-sectional incidence of tariff shocks. The response rises from zero over the first half-year, and at the reported horizon it is 1.012 (0.261) at $\rho = 1$ and 0.929 (0.245) at $\hat{\rho}$, statistically indistinguishable from full pass-through in both cases, so the model-implied response matches the magnitude of the realized consumer-price response and not only its sign.

Figure 5 also shows that the PCE pass-through estimates are driven almost entirely by the final-import component of tariff exposure. Replacing total exposure with final-import exposure produces very similar dynamic coefficients. This implies that we cannot take Figure 5 as an “out-of-sample” test for how well the model-implied domestic-production price fits the data. However, the result should not be read as evidence that the domestic-production channel is negligible either. It instead indicates that the PCE specification has limited independent variation with which to separately identify the domestic-production channel.

There is a simple reason for this. The final-import component preserves the sharp cross-sectional differences in final import content across expenditure categories. The domestic-production component, by contrast, is filtered through the domestic input-output network and then mapped into aggregate NIPA categories, after being shrunk further by retailers, wholesalers, and transportation margins. This procedure is economically important for measuring total exposure, but it also compresses the relevant cross-sectional variation. For this reason, producer-price data, which are closer to the domestic production network and available at a less aggregated industry level, provide a more suitable setting for assessing the domestic-production channel.

6 From Pass-through to the Contribution to PCE Inflation

The estimates in the last section exploited cross-sectional variation. By themselves, they do not answer the question of how much tariffs change map to aggregate PCE inflation. To do so, we need to multiply them by the exposure to the 2025 tariffs. We split the contribution into the three channels of Section 4.1: the final-import channel, goods bought finished from abroad, whose exposure does not depend on ρ and only requires knowledge of the final-import exposure coefficient β_F , the network-adjusted cost response, evaluated at $\rho = 1$, and the foreign-competition channel that estimates $\rho < 1$.

Our local projection estimates, β_h , answers the following question: “What happens to prices over h months if the tariff rises once and stays?”. In other words, it considers the tariff change as permanent. In what follows, we consider the effect of tariffs between January and

October 2025.¹⁷

Figure 6 reports the aggregate number also splitting it into the three different channels. Over the analyzed time frame, the contribution sum to 0.93 percentage points of headline PCE. Final imports account for more than half of the aggregate PCE effect. Network adjusted costs, on the other hand, account for around one third of the total predicted PCE effect. The foreign competition channel added 0.11, not negligible but smaller in magnitude than the other two channels.

Figure 7 to Figure 10 splits the aggregate number across different dimensions.

Figure 7 splits it by the month in which the tariff changed. It shows that the bulk of the total effect was concentrated between March and May, with some other relatively small increases from July to October.

Figure 8 splits the aggregate number by the HS2 chapter the duty was levied on. The top 3 commodities were Electrical Machinery, Vehicles, and Machinery. For the majority of commodities, the final imports channel was their more important contribution to PCE. Some exceptions to these were Iron and Steel Articles and Aluminum, where the network-adjusted costs channel was more relevant. The importance of foreign competition was also non-negligible across the board. For instance, this channel accounted for around a quarter of the tariffs contribution effect to PCE of Vehicles commodities.

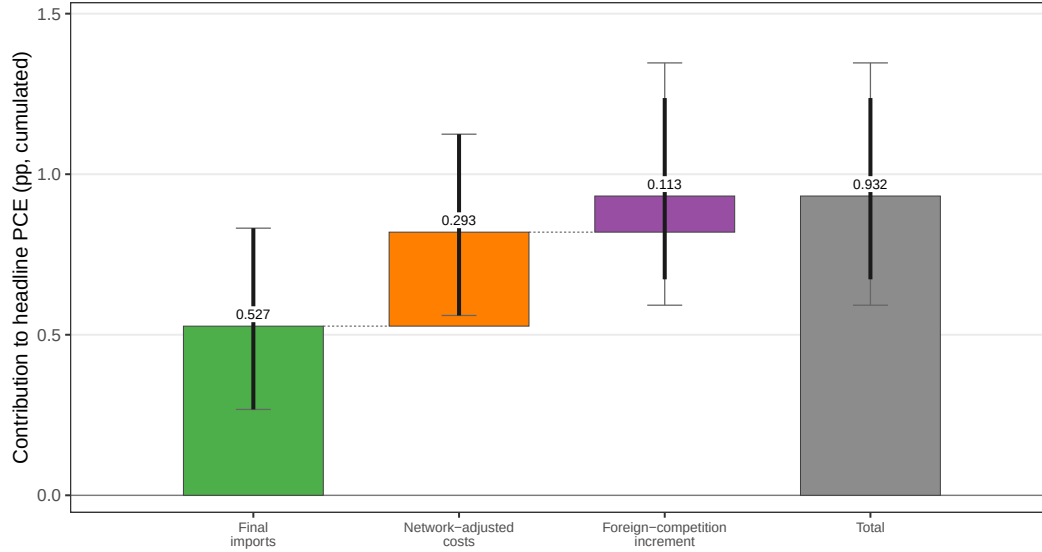
Figure 9 follows the same format as Figure 8. Half of the contribution attributed to Canada, Brazil, and Germany is due to network-adjusted costs rather than final imports, against a fifth for Vietnam and a quarter for China, whose 0.37 percentage points are three-quarters due to final imports. This highlights the importance of source-region-specific exposure.

Finally, Figure 10 shows the results by consumption categories. The largest categories are goods bought finished from abroad, and the final-import segment dominates them: garments (0.08 percentage points) and video and audio equipment (0.05) are almost entirely final imports. The exceptions are new motor vehicles (0.09, the largest category) and food purchased for off-premises consumption (0.08), which split evenly between final imports and network-adjusted costs. The services bars, on the other hand, carry no final-import segment at all: hospital and nursing home services (0.04), outpatient services (0.02), motor vehicle services, communication, and household utilities reach the consumer only through the domestic network. The largest bar is Other: twenty-eight categories individually below one percent of the total, contributing 0.11 percentage points between them, as much as the entire foreign-competition increment. Services and Other carry almost none of the cross-sectional

¹⁷We do for identification purposes: At horizon $H = 12$ the last treatment month in the sample is August 2025, so the estimates is almost entirely identified out off tariff rises.

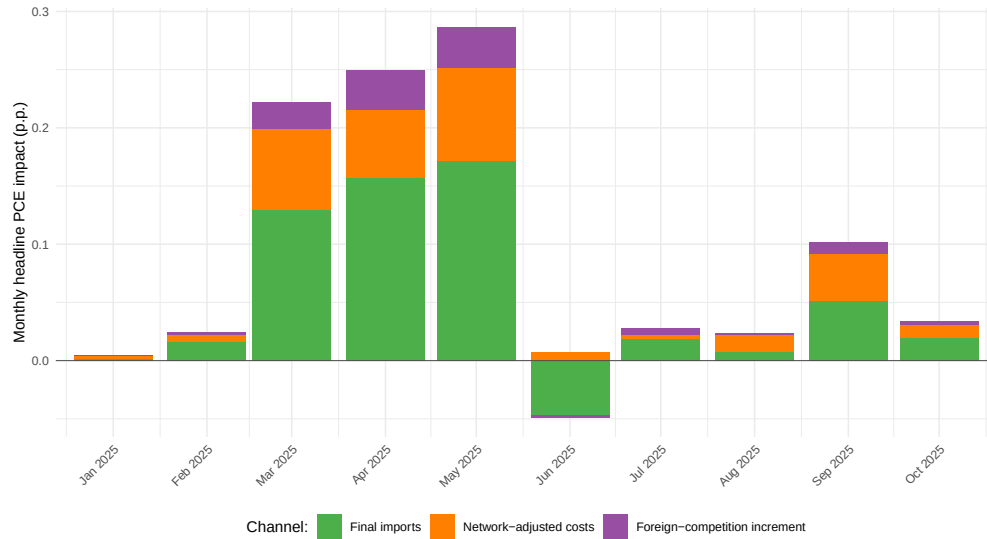
variation identifying β_F —the limited independent variation of Section 5.2—yet they account for a quarter of the contribution.

Figure 6: Contribution of the January–October 2025 Tariff Increases to Headline PCE



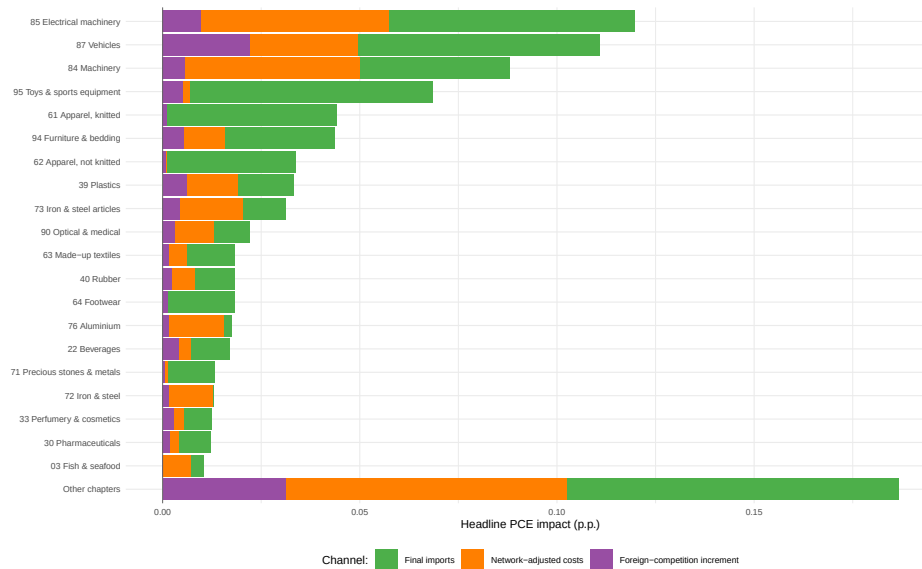
Notes: First-order contribution of the tariff increases enacted January–October 2025 to headline PCE inflation, in percentage points. The bars accumulate left to right: final imports, the network-adjusted cost response at $\rho = 1$, and the increment from setting $\rho = \hat{\rho}$. Estimates under level pass-through (not rejected by either PPI or PCE estimations). Exposure from Detail BEA commodities with commodity-specific source-region import weights. Thick whiskers propagate the 95% interval of the final-import coefficient; thin capped whiskers additionally sweep ρ over its identification-robust set.

Figure 7: Contribution by Month of Tariff Change



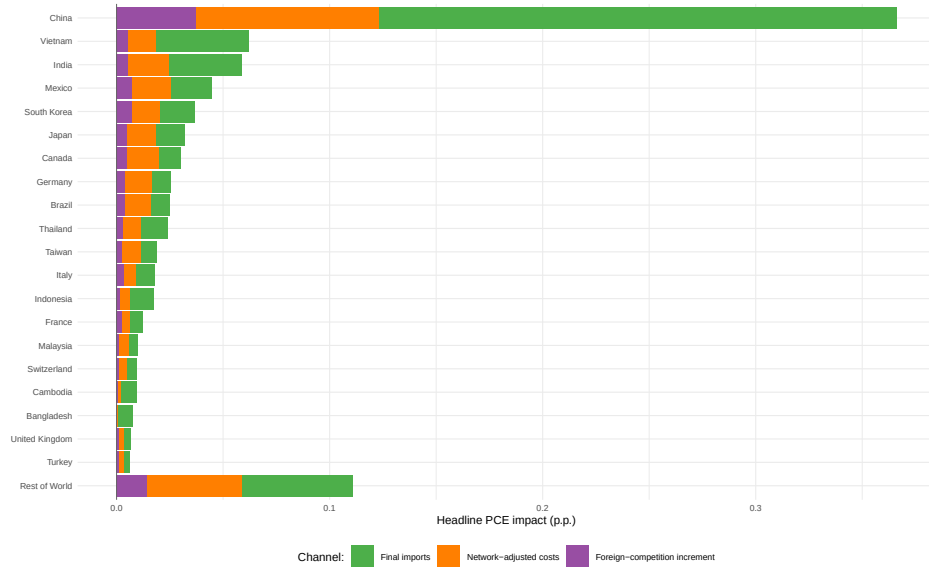
Notes: The headline PCE contribution of Figure 6, in percentage points, split by the calendar month in which the tariff changed. Each bar stacks that month's three channels, so the bars sum to the total by construction.

Figure 8: Contribution by Tariffed Commodity (HS2 Chapter)



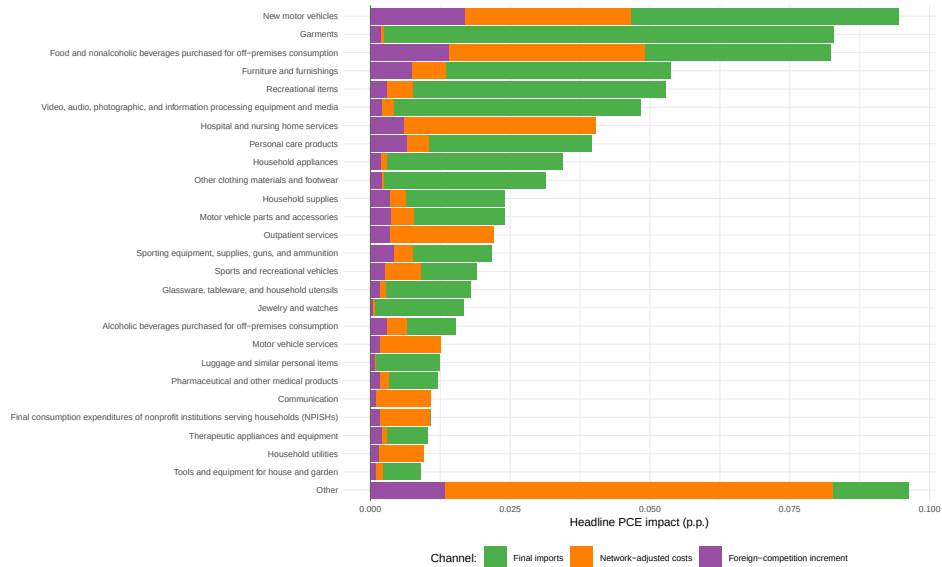
Notes: The headline PCE contribution of Figure 6, in percentage points, split by the HS2 chapter on which the tariff was levied. Each bar stacks that chapter's three channels, so the bars sum to the total by construction. The twenty chapters with the largest contribution in absolute value are shown separately and the remainder is collected in *Other chapters*. Negative segments are reported as such.

Figure 9: Contribution by Source Country



Notes: The headline PCE contribution of Figure 6, in percentage points, split by country of origin. Each bar stacks that country's three channels, so the bars sum to the total by construction and not to tariff rates. The twenty countries with the largest contribution are shown separately and the remainder is collected in *Rest of World*.

Figure 10: Contribution by Consumption Category



Notes: The headline PCE contribution of Figure 6, in percentage points, split by the NIPA category on which the affected price change is spent rather than by where the tariff was levied. Each bar stacks that category's three channels, so the bars sum to the total by construction. Categories accounting for less than one percent of the total absolute contribution are collected in *Other*.

7 Conclusion

This paper develops and estimates a framework that maps border price shocks through domestic production networks and oligopolistic pricing into producer and consumer prices. Applied to the 2025 US tariffs, the data confirm essentially complete pass-through at the border, but a model in which tariffs affect domestic producers only through their own costs substantially understates the subsequent PPI response even after input-output propagation is taken into account. Allowing domestic firms to respond to higher prices of competing imported varieties closes most of this gap: estimating the strength of this channel from the scale and cross-sectional pattern of producer-price responses gives $\hat{\rho} = 0.655$, and the resulting model is consistent with both producer- and consumer-price responses. The result is robust to allowing for within-industry heterogeneity in exposure and pass-through, while the 2018–19 episode is too weakly identified to provide an independent estimate of the competition parameter. Because the framework maps the tariff schedule into the full PCE expenditure system, the same estimated model can also be aggregated using observed expenditure weights. On this basis, the January–October 2025 tariff increases, treated as permanent, imply a first-order contribution of 0.93 percentage points to headline PCE inflation, including 0.11 percentage points from the oligopolistic competition channel.

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A Model Details

A.1 Price indices

A.1.1 Intermediate input price indices

$$p_{fi}^x(\{p_{fi,j}^x\}_{j \in \mathcal{I}}) = \min_{\{x_{fi,j}\}_{j \in \mathcal{I}}} \left\{ \sum_{j \in \mathcal{I}} p_{fi,j}^x x_{fi,j} \quad \text{s.t.} \quad x_{fi}(\{x_{fi,j}\}_{j \in \mathcal{I}}) \geq 1 \right\}, \quad (\text{A.1})$$

$$p_{fi,j}^x(\{p_{gj}\}_{g \in \mathcal{D}_j}) = \min_{\{x_{fi,gj}\}_{g \in \mathcal{D}_j}} \left\{ \sum_{g \in \mathcal{D}_j} p_{gj} x_{fi,gj} \quad \text{s.t.} \quad x_{fi,j}(\{x_{fi,gj}\}_{g \in \mathcal{D}_j}) \geq 1 \right\}, \quad (\text{A.2})$$

Both aggregators define price indices as a function of firms' prices:

$$p_{fi,j}^x = p_{fi,j}^x(\{p_{gj}\}_{g \in \mathcal{D}_j}), \quad (\text{A.3})$$

which is the price paid by firm f in industry i for inputs from j and

$$p_{fi}^x = p_{fi}^x(\{p_{fi,j}^x\}_{j \in \mathcal{I}}), \quad (\text{A.4})$$

is the *domestic* intermediate input bundle paid by firm f in industry i .

Up to a first-order approximation both price indices (equations A.3 and A.4) satisfies

$$d \log p_{fi,j}^x = \sum_{g \in \mathcal{D}_j} \frac{p_{gj} x_{fi,gj}}{p_{fi,j}^x x_{fi,j}} d \log p_{gj} = \sum_{g \in \mathcal{D}_j} \omega_{fi,gj} d \log p_{gj} \quad (\text{A.5})$$

$$d \log p_{fi}^x = \sum_{j \in \mathcal{I}} \frac{p_{fi,j}^x x_{fi,j}}{p_{fi}^x x_{fi}} d \log p_{fi,j}^x = \sum_{j \in \mathcal{I}} \omega_{fi,j} d \log p_{fi,j}^x \quad (\text{A.6})$$

In the main text, we showed the assumption we needed to empirically implement our model with the available data. We repeat the assumption here so that this appendix is self-contained.

Assumption A.1 (Aggregation of input expenditure shares). *For every pair of industries i, j ,*

$$\begin{aligned} \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j} \omega_{fi,gj} &= \rho_i^d \Omega_{i,j} s_{gj}^d \quad \text{for every } g \in D_j, \\ \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j}^* &= \rho_i^d \Omega_{i,j}^*, \end{aligned}$$

where

$$\rho_i^d \equiv \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi}.$$

Here, $\Omega_{fi,j}$ and $\Omega_{fi,j}^*$ are expenditures on domestic and imported inputs from industry j as shares of firm fi 's total variable costs, and $\omega_{fi,gj}$ is supplier gj 's share of firm fi 's expenditure on domestic inputs from industry j . The corresponding industry-level total-variable-cost shares are $\Omega_{i,j}$ and $\Omega_{i,j}^*$. All shares and pass-through coefficients are evaluated at the initial equilibrium.¹⁸

Assumption A.1 allows input expenditure shares and marginal-cost changes to differ across firms, but still allows us to use the producer price index for other industries as the relevant price that enters marginal cost calculations. Holding wages and productivity fixed, each firm's marginal-cost change satisfies

$$d \log mc_{fi} = \sum_j \Omega_{fi,j} \sum_{g \in D_j} \omega_{fi,gj} d \log p_{gj} + \sum_j \Omega_{fi,j}^* d \log p_j^*.$$

Multiplying by $\tilde{\delta}_{fi} \rho_{fi}$ and summing across domestic firms yields

$$\begin{aligned} & \sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi} \\ &= \sum_j \sum_{g \in D_j} \left(\sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j} \omega_{fi,gj} \right) d \log p_{gj} + \sum_j \left(\sum_{f \in D_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j}^* \right) d \log p_j^* \\ &= \rho_i^d \left(\sum_j \Omega_{i,j} d \log p_j^d + \sum_j \Omega_{i,j}^* d \log p_j^* \right), \end{aligned}$$

where the last equality uses the assumption and the definition $d \log p_j^d = \sum_{g \in D_j} s_{gj}^d d \log p_{gj}$.

Thus, the marginal-cost term entering the industry pricing equation is unchanged, even though individual firms need not experience the same marginal-cost change.

A.2 Channels of Sectoral PPI Changes

We provide more details about equation (37) in Proposition 3.

Equation (37) shows how domestic price indices react to a change in the foreign price $d \log p_{Fi}$ and firms' marginal costs $d \log mc_{fi}$. We can rewrite it as¹⁹:

¹⁸If a firm purchases no domestic inputs from industry j , its joint expenditure share $\Omega_{fi,j} \omega_{fi,gj}$ is defined to be zero.

¹⁹We approximate this equation under the assumption that $\pi_{Fi} \approx s_{Fi}$ and $\tilde{\pi}_{fi} \approx \tilde{\delta}_{fi}$.

$$d \log p_i^d = \frac{v_i^d (\rho_i^d d \log mc_i + \text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i)) + (1 - \rho_i^d) s_{Fi} d \log p_{Fi}}{\rho_i^d + (1 - \rho_i^d) s_{Fi}}, \quad (\text{A.7})$$

where $d \log mc_i = \mathbb{E}_{\tilde{\delta}_i}[d \log \mathbf{mc}_i]$, $\rho_i^d = \mathbb{E}_{\tilde{\delta}_i}[\rho_i]$. Note we also introduce expectation operators such that $\mathbb{E}_\omega[x] = \sum_i \omega_i x_i$ and $\text{Cov}_\omega[x, y] = E_\omega[xy] - \mathbb{E}_\omega[x] \mathbb{E}_\omega[y]$.

There are three channels that can generate a wedge between industry-level price changes $d \log p_i^d$ and marginal cost changes $d \log \mathbf{mc}_i$:

1. **Markup assumption (levels vs logs).** Captured by ν_i^d , which attenuates the pass-through of marginal cost shocks.
2. **Foreign-competition channel.** When $0 < \rho_i^d < 1$, firms respond to movements in the reference price, which itself depends on foreign competitors through s_{Fi} .
3. **Within-industry exposure heterogeneity.** Captured by the covariance term $\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i)$, which reflects the interaction between firm-level pass-through and firm-level exposure to shocks.

The covariance term is the analogue of the aggregate markup-adjustment term in [Amiti, Itskhoki and Konings \(2019\)](#). In their exchange-rate setting, firms differ both in their exposure to the exchange-rate shock and in the extent to which they absorb cost shocks through markups. In our setting, we condition directly on border price changes and decompose firms' exposure into imported-input marginal-cost exposure and foreign-competitor price exposure. The covariance term captures whether the domestic firms most exposed to the foreign cost shock are also the firms with high pass-through.²⁰

Since we already analyzed both the markup assumption and the foreign-competition channels in the main text, in this appendix we study in more detail the covariance channel. The goal of this is to be able to take this channel to industry-level data without access to firm-level data. It turns out this boils down to imposing assumptions on a covariance matrix that reduces the number of parameters to be estimated.

A.2.1 Covariance channel

We approximate firm-level marginal cost of firms within an industry as

$$d \log mc_{fi} = \sum_j \Omega_{fi,j}^d d \log p_j^d + \sum_j \Omega_{fi,j}^* d \log p_j^*, \quad (\text{A.8})$$

²⁰ [Amiti, Itskhoki and Konings \(2019\)](#) write the covariance in terms of markup adjustment, whereas we write it in terms of own-cost pass-through. Since these objects are complements, the sign convention differs mechanically.

Using (A.8), the covariance term can be written as

$$\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i) = \sum_j \text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, \boldsymbol{\Omega}_{i,j}^d) d \log p_j^d + \sum_j \text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, \boldsymbol{\Omega}_{i,j}^*) d \log p_j^* \quad (\text{A.9})$$

$$= \sum_j \mathbb{C}_{ij}^d d \log p_j^d + \sum_j \mathbb{C}_{ij}^* d \log p_j^*, \quad (\text{A.10})$$

where

$$\mathbb{C}_{ij}^* = \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} (\rho_{fi} - \rho_i^d) \Omega_{fi,j}^*. \quad (\text{A.11})$$

Define the exposure-weighted pass-through among firms exposed to input j :

$$\rho_{ij}^* \equiv \frac{\sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j}^*}{\sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \Omega_{fi,j}^*} = \frac{\sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} \Omega_{fi,j}^*}{\Omega_{ij}^*}. \quad (\text{A.12})$$

Then (A.11) can be rewritten as

$$\mathbb{C}_{ij}^* = \Omega_{ij}^* (\rho_{ij}^* - \rho_i^d). \quad (\text{A.13})$$

Restrictions. We now impose the following identification assumptions:

1. *Within industry i , the difference between exposure-weighted pass-through and average pass-through is constant across input sectors:*

$$\rho_{ij}^* - \rho_i^d = \gamma_i \quad \forall j. \quad (\text{A.14})$$

2. $\mathbb{C}^d = \mathbf{0}$

Under the first assumption, (A.13) simplifies to

$$\mathbb{C}_{ij}^* = \gamma_i \Omega_{ij}^*. \quad (\text{A.15})$$

The first restriction implies that within industry i , firms that are more exposed to imported inputs systematically have pass-through that differs from the industry average by a constant amount γ_i , independently of which input sector generates the exposure.

When $\gamma_i > 0$, firms that are more exposed to import costs also tend to have higher pass-through, amplifying the effect of foreign shocks. When $\gamma_i < 0$, the opposite holds. This is simply a plausible assumption to allow for limited degrees of freedom in the nonparametric

estimation.²¹ The second restriction is also necessary to reduce degrees of freedom. Since $\mathbb{C}^d$ affects prices indirectly, this assumption should not have a large impact.

Under such restrictions equation (A.7), in vector form, becomes

$$d \log \mathbf{p}^d = \left[I - (I - \boldsymbol{\rho}^d)(I - \mathbf{S}_F) - \boldsymbol{\rho}^d \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^d \right]^{-1} \left[(\boldsymbol{\rho}^d + \boldsymbol{\gamma}) \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^* + (I - \boldsymbol{\rho}^d) \mathbf{S}_F \right] d \log \mathbf{p}^* \quad (\text{A.16})$$

Equation (A.16) implies that a covariance term would possibly amplify or attenuate the effects of marginal costs increases, in accordance to an estimated diagonal matrix $\boldsymbol{\gamma}$ that summarizes how the distribution of markups and affected firms within the industries are affected by the tariffs. Effectively, under our restriction assumption, the matrix $\boldsymbol{\gamma}$ can further disentangle the link between predicted and observed marginal costs.

The issue with this equation is that both the matrix $\boldsymbol{\rho}$ and $\boldsymbol{\gamma}$ are unobserved. Therefore this is a system with N equations and $2N$ parameters. In practice, we aim at parameterizing such values with industry observable characteristics such as HHI index, markup size, foreign shares, and others. Of course equation (A.16) also implies that any model estimating ρ without accounting for the potential impact of γ would be biased.

Note that this channel cannot be active in the intermediate sector unless $\boldsymbol{\rho}^d \neq I$. Under the normalization $\boldsymbol{\rho}^d = I$, the covariance wedge $\mathbb{C}^*$ collapses to zero. This is because if the average ρ_i^d in an industry is 1, then all $\rho_{fi} = 1$ for all firms f in i . But this implies that the covariance term between ρ_{fi} and the marginal cost changes is zero.

Retail-side covariance wedge. A related covariance wedge may arise in the final-demand layer. Unlike domestic producers, retailers do not compete with foreign retailers. Imported finished goods have already crossed the border when they enter this layer and are purchased as inputs by domestic retailers. There is therefore no retail analogue of the foreign-competition channel governed by $\boldsymbol{\rho}^d$. Retailers may nevertheless differ both in their own-cost pass-through and in their exposure to domestic and imported commodities.

To make this distinction explicit, recall from the main text that retailer-level marginal-cost changes satisfy

$$d \log mc_{fe} = \sum_j \theta_{fe,j}^d d \log p_j^d + \sum_j \theta_{fe,j}^* d \log p_j^*. \quad (\text{A.17})$$

Under the approximation $\pi_{fe} \approx \delta_{fe}$, define the cost-weighted average own-cost pass-through within expenditure category e as

$$\rho_e^r \equiv \sum_{f \in e} \delta_{fe} \rho_{fe}. \quad (\text{A.18})$$

²¹We have also allowed for $\mathbb{C}$ to be a free diagonal matrix and our non-parametric estimation has shown that the cost share is strongly correlated to the diagonal elements.

Corollary 1 can then be rewritten as

$$\mathrm{d} \log p_e = v_e \left[\mathrm{d} \log \mathbf{mc}_e + \frac{\mathbb{Cov}_{\delta_e}(\boldsymbol{\rho}_e, \mathrm{d} \log \mathbf{mc}_e)}{\rho_e^r} \right]. \quad (\text{A.19})$$

Using equation (A.17), define the two retail-side covariance matrices by

$$\mathbb{C}_{ej}^{r,d} \equiv \frac{\mathbb{Cov}_{\delta_e}(\boldsymbol{\rho}_e, \boldsymbol{\theta}_{e,j}^d)}{\rho_e^r}, \quad (\text{A.20})$$

$$\mathbb{C}_{ej}^{r,*} \equiv \frac{\mathbb{Cov}_{\delta_e}(\boldsymbol{\rho}_e, \boldsymbol{\theta}_{e,j}^*)}{\rho_e^r}. \quad (\text{A.21})$$

The general retail-price equation is therefore

$$\mathrm{d} \log \mathbf{p} = \boldsymbol{\Upsilon}^r \left[(\boldsymbol{\Theta}^d + \mathbb{C}^{r,d}) \mathrm{d} \log \mathbf{p}^d + (\boldsymbol{\Theta}^* + \mathbb{C}^{r,*}) \mathrm{d} \log \mathbf{p}^* \right]. \quad (\text{A.22})$$

Let $\boldsymbol{\Phi}^{d,*}$ denote the mapping from imported prices into domestic producer prices under the chosen producer-side specification:

$$\mathrm{d} \log \mathbf{p}^d = \boldsymbol{\Phi}^{d,*} \mathrm{d} \log \mathbf{p}^*. \quad (\text{A.23})$$

Under the baseline specification of full pass-through and no covariance wedge,

$$\boldsymbol{\Phi}^{d,*} = (\mathbf{I} - \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^d)^{-1} \boldsymbol{\Upsilon}^d \boldsymbol{\Omega}^*. \quad (\text{A.24})$$

Substituting equation (A.23) into equation (A.22) yields

$$\mathrm{d} \log \mathbf{p} = \boldsymbol{\Upsilon}^r \left[(\boldsymbol{\Theta}^d + \mathbb{C}^{r,d}) \boldsymbol{\Phi}^{d,*} + \boldsymbol{\Theta}^* + \mathbb{C}^{r,*} \right] \mathrm{d} \log \mathbf{p}^*. \quad (\text{A.25})$$

Equation (A.25) makes clear that retailer heterogeneity can affect both the domestic-production channel, through $\mathbb{C}^{r,d}$, and the final-import channel, through $\mathbb{C}^{r,*}$.

Final-import restriction. The two retail-side covariance matrices cannot be estimated unrestrictedly from expenditure-category price variation alone. As a baseline empirical specification, we therefore impose

$$\mathbb{C}^{r,d} = \mathbf{0}, \quad \mathbb{C}^{r,*} = \boldsymbol{\Gamma}^r \boldsymbol{\Theta}^*, \quad \boldsymbol{\Gamma}^r = \text{diag}(\{\gamma_e^r\}_{e \in \mathcal{E}}). \quad (\text{A.26})$$

Under this restriction, equation (A.25) becomes

$$d \log \mathbf{p} = \Upsilon^r [\Theta^d \Phi^{d,*} + (\mathbf{I} + \Gamma^r) \Theta^*] d \log \mathbf{p}^*. \quad (\text{A.27})$$

This restriction is not a theoretical implication of the model. It is a parsimonious normalization that isolates the covariance between retailer-level own-cost pass-through and final-import exposure, which is the component that generates the sharpest cross-sectional variation across expenditure categories.

Alternative sourcing-substitution specification. As an alternative, suppose that retailers within an expenditure category differ in whether they source a given commodity domestically or internationally, but not in their total exposure to that commodity:

$$\theta_{fe,j}^d + \theta_{fe,j}^* = \theta_{e,j} \quad \text{for all } f \in e. \quad (\text{A.28})$$

Under this assumption,

$$\mathbb{C}^{r,d} = -\mathbb{C}^{r,*}. \quad (\text{A.29})$$

Combining equation (A.29) with the proportionality restriction

$$\mathbb{C}^{r,*} = \Gamma^r \Theta^* \quad (\text{A.30})$$

gives

$$\begin{aligned} d \log \mathbf{p} &= \Upsilon^r [(\Theta^d - \Gamma^r \Theta^*) \Phi^{d,*} + (\mathbf{I} + \Gamma^r) \Theta^*] d \log \mathbf{p}^* \\ &= \Upsilon^r [\Theta^d \Phi^{d,*} + \Theta^* + \Gamma^r \Theta^* (\mathbf{I} - \Phi^{d,*})] d \log \mathbf{p}^*. \end{aligned} \quad (\text{A.31})$$

This alternative specification gives Γ^r a different interpretation. Retailers with higher own-cost pass-through are disproportionately exposed to imported rather than domestically sourced versions of the same commodities. The same wedge therefore raises the final-import contribution and correspondingly lowers the contribution of domestic sourcing.

B Proofs

B.1 Proof of Proposition 1

Start from the pricing condition

$$p_{fi} = mc_{fi} + \frac{\sigma_i}{1 - \pi_{fi}}. \quad (\text{B.1})$$

Differentiating this expression

$$dp_{fi} = dmc_{fi} + \frac{\mu_{fi}}{1 - \pi_{fi}} d\pi_{fi}.$$

Differentiating the share, we get

$$\begin{aligned} d\pi_{fi} &= -\frac{\pi_{fi}}{\sigma_i} \left(dp_{fi} - \sum_{g \in i} \pi_{gi} dp_{gi} \right) \\ d\pi_{fi} &= -\frac{\pi_{fi}}{\sigma_i} (1 - \pi_{fi}) \left(dp_{fi} - \sum_{g \in i \setminus f} \tilde{\pi}_{gi} dp_{gi} \right) \\ d\pi_{fi} &= -\frac{\pi_{fi}}{\sigma_i} (1 - \pi_{fi}) (dp_{fi} - dp_{-f,i}), \end{aligned}$$

where we defined the competitors' price index of firm f as

$$dp_{-f,i} = \sum_{g \in i \setminus f} \tilde{\pi}_{gi} dp_{gi}, \quad (\text{B.2})$$

with $\tilde{\pi}_{gi} = \frac{\pi_{gi}}{\sum_{h \in i \setminus f} \pi_{hi}}$.

Replacing into the differentiated pricing condition

$$\begin{aligned} dp_{fi} &= dmc_{fi} - \frac{\mu_{fi}}{1 - \pi_{fi}} \frac{\pi_{fi}}{\sigma_i} (1 - \pi_{fi}) (dp_{fi} - dp_{-f,i}) \\ dp_{fi} &= dmc_{fi} - \frac{\pi_{fi}}{1 - \pi_{fi}} (dp_{fi} - dp_{-f,i}) \end{aligned}$$

Rearranging, we get

$$dp_{fi} = (1 - \pi_{fi}) dmc_{fi} + \pi_{fi} dp_{-f,i}, \quad (\text{B.3})$$

which is the expression in the main text.

B.2 Proof of Proposition 2

Divide the final expression of Proposition 1 by p_{fi} and rearrange to get

$$d \log p_{fi} = (1 - \pi_{fi}) \frac{mc_{fi}}{p_{fi}} d \log mc_{fi} + \pi_{fi} \frac{p_{-f,i}}{p_{fi}} d \log p_{-f,i}. \quad (\text{B.4})$$

B.3 Proof of Proposition 3

Start from the definition of the producer price index

$$\begin{aligned} d \log p_i^d &= \sum_{f \in \mathcal{D}_i} s_{fi}^d d \log p_{fi} \\ d \log p_i^d &= \sum_{f \in \mathcal{D}_i} s_{fi}^d (1 - \alpha_{fi}) \left[\rho_{fi} d \log mc_{fi} + (1 - \rho_{fi}) \frac{R_i}{mc_{fi} q_i} d \log p_i \right] \\ d \log p_i^d &= \left(\sum_{g \in \mathcal{D}_i} s_{gi}^d (1 - \alpha_{gi}) \right) \left(\sum_{g \in \mathcal{D}_i} \frac{s_{gi}^d (1 - \alpha_{gi})}{\left(\sum_{g \in \mathcal{D}_i} s_{gi}^d (1 - \alpha_{gi}) \right)} \rho_{gi} \right) \sum_{f \in \mathcal{D}_i} \frac{s_{fi}^d (1 - \alpha_{fi}) \rho_{fi}}{\sum_{g \in \mathcal{D}_i} s_{gi}^d (1 - \alpha_{gi}) \rho_{gi}} d \log mc_{fi} \\ &\quad + \left[\sum_{f \in \mathcal{D}_i} \frac{s_{fi}^d}{mc_{fi}} (1 - \alpha_{fi}) (1 - \rho_{fi}) \right] \frac{R_i}{q_i} d \log p_i \\ d \log p_i^d &= \frac{VC_i^d}{R_i^d} \rho_i^d \sum_{f \in \mathcal{D}_i} \gamma_{fi} d \log mc_{fi} + \left[\sum_{f \in \mathcal{D}_i} \frac{s_{fi}^d}{mc_{fi}} (1 - \alpha_{fi}) (1 - \rho_{fi}) \right] \frac{R_i}{q_i} d \log p_i \\ d \log p_i^d &= \frac{VC_i^d}{R_i^d} \rho_i^d \sum_{f \in \mathcal{D}_i} \gamma_{fi} d \log mc_{fi} + \frac{R_i}{R_i^d} \left(\sum_{f \in \mathcal{D}_i} \pi_{fi} (1 - \rho_{fi}) \right) d \log p_i \\ d \log p_i^d &= \frac{VC_i^d}{R_i^d} \rho_i^d \sum_{f \in \mathcal{D}_i} \gamma_{fi} d \log mc_{fi} + \frac{R_i}{R_i^d} (1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi} (1 - \rho_{fi}) \right) d \log p_i \end{aligned}$$

We first solve for $d \log p_{fi}$ as a function of marginal costs and the industry price index. To do so, recall that the level change in the markup is

$$d \mu_{fi} = -\mu_{fi} \frac{\pi_{fi}}{\sigma_i (1 - \pi_{fi})} \left(dp_{fi} - \sum_{g \in i} \pi_{gi} dp_{gi} \right) = -\frac{\pi_{fi}}{(1 - \pi_{fi})^2} (dp_{fi} - dp_i^{eff}),$$

where we use the definition of the change in the effective price index

$$dp_i^{eff} = \sum_{g \in i} \pi_{gi} dp_{gi}. \quad (\text{B.5})$$

Replacing the change in the markup into the pricing condition

$$\begin{aligned}
dp_{fi} &= dmc_{fi} + d\mu_{fi} \\
dp_{fi} &= dmc_{fi} - \frac{\pi_{fi}}{(1 - \pi_{fi})^2} (dp_{fi} - dp_i^{eff}) \\
dp_{fi} &= \frac{1}{1 + \frac{\pi_{fi}}{(1 - \pi_{fi})^2}} dmc_{fi} + \frac{\frac{\pi_{fi}}{(1 - \pi_{fi})^2}}{1 + \frac{\pi_{fi}}{(1 - \pi_{fi})^2}} dp_i^{eff}.
\end{aligned}$$

This implies

$$dp_{fi} = \rho_{fi} dmc_{fi} + (1 - \rho_{fi}) dp_i^{eff}, \quad (\text{B.6})$$

where

$$\rho_{fi} = \frac{1}{1 + \frac{\pi_{fi}}{(1 - \pi_{fi})^2}} = \frac{(1 - \pi_{fi})^2}{(1 - \pi_{fi})^2 + \pi_{fi}} = \frac{(1 - \pi_{fi})^2}{1 - \pi_{fi} + \pi_{fi}^2}$$

We can link the effective price index to the industry price index change $d \log p_i$:

$$\begin{aligned}
dp_i^{eff} &= \sum_{f \in i} p_{fi} \pi_{fi} d \log p_{fi}, \\
&= \sum_{f \in i} \frac{p_{fi} q_{fi}}{q_i} d \log p_{fi}, \\
&= \frac{R_i}{q_i} \sum_{f \in i} \frac{p_{fi} q_{fi}}{R_i} d \log p_{fi}, \\
dp_i^{eff} &= \frac{R_i}{q_i} d \log p_i.
\end{aligned}$$

Using this equality and expressing the pricing condition in percentage terms

$$\begin{aligned}
dp_{fi} &= \rho_{fi} dmc_{fi} + (1 - \rho_{fi}) \frac{R_i}{q_i} d \log p_i, \\
d \log p_{fi} &= \rho_{fi} (1 - \alpha_{fi}) d \log mc_{fi} + (1 - \rho_{fi}) \frac{R_i}{p_{fi} q_i} d \log p_i.
\end{aligned}$$

Multiplying by s_{fi}^d and adding across domestic firms we get

$$\begin{aligned}
d \log p_i^d &= \sum_{f \in \mathcal{D}_i} s_{fi}^d \left(\rho_{fi}(1 - \alpha_{fi}) d \log mc_{fi} + (1 - \rho_{fi}) \frac{R_i}{p_{fi} q_i} d \log p_i \right) \\
d \log p_i^d &= \sum_{f \in \mathcal{D}_i} s_{fi}^d \rho_{fi}(1 - \alpha_{fi}) d \log mc_{fi} + \sum_{f \in \mathcal{D}_i} s_{fi}^d (1 - \rho_{fi}) \frac{R_i}{p_{fi} q_i} d \log p_i \\
d \log p_i^d &= \sum_{f \in \mathcal{D}_i} \frac{p_{fi} q_{fi}}{R_i^d} \rho_{fi}(1 - \alpha_{fi}) d \log mc_{fi} + \sum_{f \in \mathcal{D}_i} \frac{p_{fi} q_{fi}}{R_i^d} (1 - \rho_{fi}) \frac{R_i}{p_{fi} q_i} d \log p_i \\
d \log p_i^d &= \sum_{f \in \mathcal{D}_i} \frac{mc_{fi} q_{fi}}{R_i^d} \rho_{fi} d \log mc_{fi} + \sum_{f \in \mathcal{D}_i} \pi_{fi}(1 - \rho_{fi}) \frac{R_i}{R_i^d} d \log p_i \\
d \log p_i^d &= \frac{VC_i^d}{R_i^d} \sum_{f \in \mathcal{D}_i} \frac{mc_{fi} q_{fi}}{VC_i^d} \rho_{fi} d \log mc_{fi} + \sum_{f \in \mathcal{D}_i} \pi_{fi}(1 - \rho_{fi}) \frac{R_i}{R_i^d} d \log p_i \\
d \log p_i^d &= v_i^d \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi} + (1 - \pi_{Fi}) \sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi}) \frac{1}{s_i^d} d \log p_i
\end{aligned}$$

Now, we use the definition of $d \log p_i = s_i^d d \log p_i^d + s_{Fi} d \log p_{Fi}$, to get

$$d \log p_i^d = v_i^d \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi} + (1 - \pi_{Fi}) \sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi}) \frac{1}{s_i^d} (s_i^d d \log p_i^d + s_{Fi} d \log p_{Fi}).$$

Rearranging, we get

$$\begin{aligned}
d \log p_i^d &= \frac{1}{\left(1 - (1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})\right)\right)} \left(v_i^d \left(\sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi} \right) \right), \quad (\text{B.7}) \\
&+ \frac{1}{\left(1 - (1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})\right)\right)} \left(\frac{(1 - \pi_{Fi}) \left(\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi})\right)}{(1 - s_{Fi})} s_{Fi} d \log p_{Fi} \right),
\end{aligned}$$

which is the expression in the main text.

B.4 Proof of Proposition 4

From the earlier section, we have an expression for the PPI change of an industry i :

$$d \log p_i^d = v_i^d \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi} d \log mc_{fi} + (1 - \pi_{Fi}) \sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi}(1 - \rho_{fi}) \frac{1}{s_i^d} (s_i^d d \log p_i^d + s_{Fi} d \log p_{Fi}) \quad (\text{B.8})$$

Given our assumptions, we have $d \log mc_{fi} = d \log mc_i$ for all $f \in \mathcal{D}_i$. Moreover, we defined:

$$\rho_i^d = \mathbb{E}_{\tilde{\delta}_i}(\rho_i) = \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi},$$

as the industry level pass-through of marginal cost to industry prices in levels, weighted according to domestic cost-shares $\tilde{\delta}_i = \{\tilde{\delta}_{fi}\}_{f \in \mathcal{D}_i}$.

Using these results, we can write equation (B.8) as

$$d \log p_i^d = v_i^d \rho_i^d d \log mc_i + (1 - \pi_{Fi})(1 - \mathbb{E}_{\tilde{\pi}_i}(\rho_i))(d \log p_i^d + (1 - s_{Fi})^{-1} s_{Fi} d \log p_{Fi}), \quad (\text{B.9})$$

which follows since $\sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi} = 1$ and $\mathbb{E}_{\tilde{\pi}_i}(\rho_i) = \sum_{f \in \mathcal{D}_i} \tilde{\pi}_{fi} \rho_{fi}$ denotes the expected marginal cost pass-through in levels when weighted according to domestic variety shares $\tilde{\pi}_i = \{\tilde{\pi}_{fi}\}_{f \in \mathcal{D}_i}$.

To arrive at the proposition in the main text, we assume that $\mathbb{E}_{\tilde{\pi}_i}(\rho_i) \approx \mathbb{E}_{\tilde{\delta}_i}(\rho_i) = \rho_i^d$. Hence, industry level marginal cost pass-through in levels when weighted according to either domestic variety shares or domestic cost shares are not too dissimilar. This assumption allows us to write:

$$d \log p_i^d = v_i^d \rho_i^d d \log mc_i + (1 - \pi_{Fi})(1 - \rho_i^d)(d \log p_i^d + (1 - s_{Fi})^{-1} s_{Fi} d \log p_{Fi}). \quad (\text{B.10})$$

Equation (B.10) in vector form is

$$d \log \mathbf{p}^d = \mathbf{\Upsilon}^d \boldsymbol{\rho}^d d \log \mathbf{mc} + (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d)(d \log \mathbf{p}^d + (\mathbf{I} - \mathbf{S}_F)^{-1} \mathbf{S}_F d \log \mathbf{p}_F), \quad (\text{B.11})$$

where

$$\mathbf{\Upsilon}^d = \text{diag}(\mathbf{v}^d); \quad \boldsymbol{\rho}^d = \text{diag}(\{\rho_i^d\}_{i \in \mathcal{I}}); \quad \mathbf{S}_F = \text{diag}(\mathbf{s}_F); \quad \mathbf{\Pi}_F = \text{diag}(\boldsymbol{\pi}_F).$$

Up to a first-order approximation, marginal cost changes are only a function of input price changes:

$$d \log mc_{fi} = \sum_j \Omega_{fi,j} d \log p_j^d + \sum_j \Omega_{fi,j}^* d \log p_j^*. \quad (\text{B.12})$$

Under Assumption A.1, this can be written omitting the firm subscript

$$d \log mc_i = \sum_j \Omega_{i,j} d \log p_j^d + \sum_j \Omega_{i,j}^* d \log p_j^*, \quad (\text{B.13})$$

or in vector form

$$\mathrm{d} \log \mathbf{m} \mathbf{c} = \mathbf{\Omega}^d \mathrm{d} \log \mathbf{p}^d + \mathbf{\Omega}^* \mathrm{d} \log \mathbf{p}^*, \quad (\text{B.14})$$

with $\mathbf{\Omega}^d = \{\Omega_{i,j}\}_{i,j \in \mathcal{I}}$ and $\mathbf{\Omega}^* = \{\Omega_{i,j}^*\}_{i,j \in \mathcal{I}}$.

Replacing (B.14) into (B.11), we get

$$\mathrm{d} \log \mathbf{p}^d = \mathbf{\Upsilon}^d \boldsymbol{\rho}^d (\mathbf{\Omega}^d \mathrm{d} \log \mathbf{p}^d + \mathbf{\Omega}^* \mathrm{d} \log \mathbf{p}^*) + (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d)(\mathrm{d} \log \mathbf{p}^d + (\mathbf{I} - \mathbf{S}_F)^{-1} \mathbf{S}_F \mathrm{d} \log \mathbf{p}_F).$$

Solving for $\mathrm{d} \log \mathbf{p}^d$ we get

$$\begin{aligned} \mathrm{d} \log \mathbf{p}^d &= (\mathbf{I} - \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^d - (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d))^{-1} \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^* \mathrm{d} \log \mathbf{p}^*, \\ &\quad + (\mathbf{I} - \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^d - (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d))^{-1} (\mathbf{I} - \mathbf{S}_F)^{-1} (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d) \mathbf{S}_F \mathrm{d} \log \mathbf{p}_F. \end{aligned}$$

Letting

$$\mathbf{\Phi}^* = \mathbf{G} \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^*, \quad (\text{B.15})$$

$$\mathbf{\Phi}^F = \mathbf{G} (\mathbf{I} - \mathbf{S}_F)^{-1} (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d) \mathbf{S}_F, \quad (\text{B.16})$$

$$\mathbf{G} = (\mathbf{I} - \mathbf{\Upsilon}^d \boldsymbol{\rho}^d \mathbf{\Omega}^d - (\mathbf{I} - \mathbf{\Pi}_F)(\mathbf{I} - \boldsymbol{\rho}^d))^{-1}, \quad (\text{B.17})$$

we arrive at

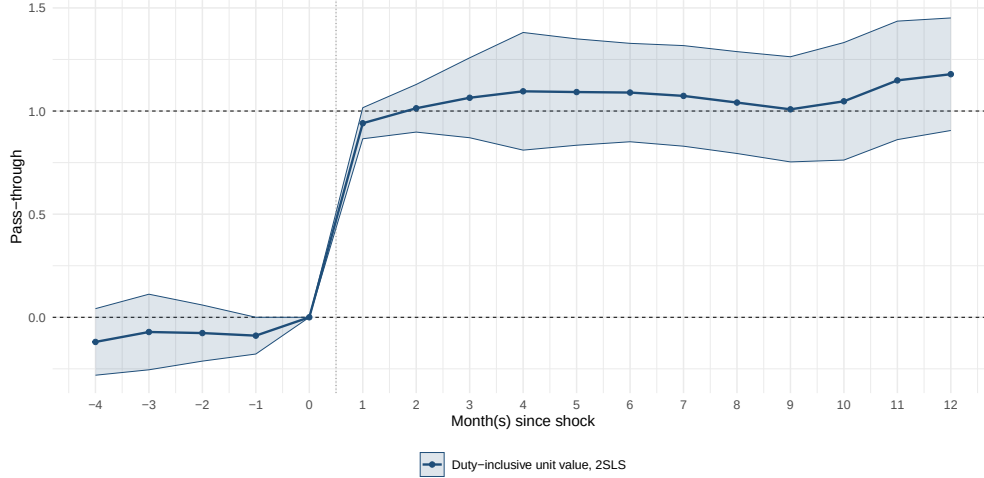
$$\mathrm{d} \log \mathbf{p}^d = \mathbf{\Phi}^* \mathrm{d} \log \mathbf{p}^* + \mathbf{\Phi}^F \mathrm{d} \log \mathbf{p}_F, \quad (\text{B.18})$$

which is the expression in the main text.

C Border Pass-through on a Balanced Panel

Figure 2 estimates each horizon on every import record whose own control block is observed, so the sample shrinks as the horizon lengthens and the lead block deepens. Here we instead require the *deepest* block — twelve statutory lags and eleven statutory leads, that is twenty-five consecutive months of trade — at every horizon, which makes the sample identical across horizons and the path pure dynamics. It costs 36 percent of the records and selects large, continuously-trading product-origin pairs. Pass-through is again complete on impact, 0.94 (0.04), and stays complete. The balanced path runs slightly above the main-text one from the second month on — by less than one standard error at every horizon — and coincides with it at the twelfth by construction.

Figure C.1: Border Pass-through of Tariffs into Import Unit Values, Balanced Panel



Notes: As Figure 2, with every horizon restricted to the records observed over the longest control block, so the sample is constant in h and the path is pure dynamics. The dashed line at one marks complete pass-through of the duty into the border price. At $h = 12$ the estimate coincides with Figure 2 by construction. Shading gives 95% confidence bands, two-way clustered by HS-8 product and country of origin.

D Data Construction and Concordances

This appendix describes the construction of the empirical objects used in the paper. Tariffs are observed at the HS-10 product-by-country level, the production network in BEA industry and commodity classifications, producer prices in NAICS-based PPI series, and consumer prices in NIPA expenditure categories. We therefore construct concordances that place the tariff shock, input-output relationships, and price outcomes on common commodity and expenditure-category indices.

For the 2018–19 episode, we use the 2017 BEA Detail input-output accounts. For the 2025 episode, we use 2024 BEA annual and Trade in Value Added (TiVA) accounts, available at the Summary level, and disaggregate them using within-block shares from the 2017 Detail Benchmark accounts. Appendix D.2 describes this procedure.

D.1 Input-output and expenditure matrices

The empirical implementation requires the domestic-input matrix Ω^d , the imported-input matrix Ω^* , the foreign-competitor share $\mathbf{S}_F$, the expenditure-category sourcing matrices Θ^d and Θ^* , and the markup-adjustment matrices Υ^d and Υ^r .

Domestic and imported intermediate inputs. The matrices Ω^d and Ω^* are defined on BEA commodities. The entry Ω_{ij}^d is the value of domestically produced commodity j used per dollar of variable cost of commodity i , while Ω_{ij}^* is the corresponding imported-input share.²² We construct both from the BEA Use and Import matrices and convert industry flows to a commodity basis using the Make matrix.

At $\rho = 1$, the network-adjusted imported-input response is

$$(\mathbf{I} - \Upsilon^d \Omega^d)^{-1} \Upsilon^d \Omega^* \mathbf{d} \log \mathbf{p}^*. \quad (\text{D.1})$$

For $\rho < 1$, the same objects enter the full producer-price mapping in equation (45).

Foreign competitors. The diagonal entry s_{Fi} of $\mathbf{S}_F$ is commodity i ’s own-market import share, measured as imports divided by domestic supply. As in the main text, we impose $\pi_{fi} = s_{fi}$ at the initial equilibrium, so observed sales shares provide the empirical counterpart of the model’s variety shares.

²²Normalising by variable cost rather than by gross output is what makes $\Upsilon^d = \mathbf{I}$ the log-pass-through case below; the product $\Upsilon^d \Omega$ is the same object under either convention.

Final-import and domestic sourcing. Let $\Theta^* = [\theta_{ej}^*]$ and $\Theta^d = [\theta_{ej}^d]$, with rows indexing NIPA expenditure categories and columns BEA commodities. We construct both from the BEA PCE bridge and commodity-level import penetration. The first records imported finished-goods content; the second records domestically produced commodity content.

At $\rho = 1$, the consumer-price sensitivity to border prices is the sum of

$$\Delta^{\text{Final}} = \Upsilon^r \Theta^* \quad (\text{D.2})$$

and

$$\Delta^{\text{Dom}} = \Upsilon^r \Theta^d (I - \Upsilon^d \Omega^d)^{-1} \Upsilon^d \Omega^*. \quad (\text{D.3})$$

The first is the final-import channel. The second captures imported inputs propagated through domestic production before reaching expenditure categories. For $\rho < 1$, the producer block is replaced by equation (45).

Markup adjustment. Under pass-through in levels,

$$\Upsilon_{ii}^d = v_i^d = \frac{VC_i^d}{R_i^d},$$

so a one-dollar increase in marginal cost passes through one for one in the price-taking limit, while its percentage effect is scaled by the initial variable-cost share. Under pass-through in logs, percentage marginal-cost changes pass through one for one and $\Upsilon^d = I$. The retail matrix Υ^r is constructed analogously from the producer-value and distribution-margin components of the PCE bridge: under levels its denominator is full purchaser value, and under logs only the variable-cost part of each distribution margin enters it.

D.2 Detail-level matrices outside Benchmark years

The BEA publishes Detail input-output tables only in Benchmark years, with 2017 the latest available for our application. Annual accounts through 2024 are available at the Summary level. For the 2025 exercise, we therefore preserve observed 2024 Summary totals while allocating each Summary cell across its Detail constituents using 2017 shares.

Let $x_{ab,2017}^D$ be a Detail cell and $X_{AB,2017}^S$ its corresponding Summary cell. Define

$$\lambda_{ab} = \frac{x_{ab,2017}^D}{X_{AB,2017}^S}.$$

The reconstructed 2024 Detail cell is

$$\hat{x}_{ab,2024}^D = \lambda_{ab} X_{AB,2024}^S.$$

The resulting Detail table aggregates exactly to the published 2024 Summary accounts. The assumption is therefore not that 2017 input-output levels remain unchanged, but only that within-Summary-block Detail shares remain fixed after 2017.

D.3 From tariff lines to BEA commodities

Customs data are observed monthly at the HS-10 product-by-country level and include a NAICS-based import classification. We first map the customs NAICS code to a BEA industry, moving to coarser NAICS levels when necessary. We then use the BEA Make matrix to convert industries into commodities.

For each episode, we construct fixed import-composition weights $a_{i,kc}$ giving the share of commodity i 's imports accounted for by HS-10 product k from country c . The commodity-level border shock is

$$d \log p_{i,t}^* = \sum_{k,c} a_{i,kc} d \log(1 + \tau_{kc,t}). \quad (\text{D.4})$$

Weights are based on 2017 trade for the 2018–19 episode and 2024 trade for the 2025 episode and are held fixed within each episode.

Approximately 95 percent of import value can be assigned to a BEA commodity. Records without a usable producing-industry classification are excluded rather than reallocated. Commodities with no matched imports remain in the system with zero border exposure.

D.4 Resolving imports by source region

Country-specific tariffs require source-region shares at the Detail commodity level. Census trade data provide detailed source composition but do not distinguish end use; the TiVA tables distinguish end use but only at the Summary level. We combine the two using iterative proportional fitting.

Let t_{jg} be region g 's share of Detail commodity j 's total imports, $B(j)$ its Summary block, and m_{Bg} region g 's published import share for that block and end use. We construct Detail

shares r_{jg} satisfying

$$\sum_g r_{jg} = 1, \quad (\text{D.5})$$

$$\sum_{j \in B} \omega_j r_{jg} = m_{Bg}, \quad (\text{D.6})$$

where ω_j is commodity j 's share of the block's import value. The solution has the form

$$r_{jg} = t_{jg} a_j b_{B(j),g}.$$

We apply this procedure separately to final imports, imported intermediates, and own-market imports. For intermediate inputs, the regional split is also allowed to vary with the using industry. By construction, aggregation back to the Summary level reproduces the published TiVA regional totals exactly.

D.5 Matching PPI series to BEA commodities

Producer-price outcomes are BLS PPI industry series, while model exposure is indexed by BEA commodity. We match the two through their common NAICS classifications.

Starting at the most detailed NAICS level, we retain a match only when the resulting PPI outcome can be assigned uniquely to one BEA commodity. Ambiguous matches are reconsidered at progressively coarser NAICS levels. When several PPI series jointly represent one commodity and no unique parent series is available, we use their unweighted average.

The procedure produces PPI outcomes for 273 BEA commodities: 252 matched to a single series and 21 to an average of multiple series.

Table D.1: Coverage of the PPI-to-BEA Match

Match	Number of BEA commodities
Single PPI series	252
Average of multiple PPI series	21
Total PPI outcomes	273

Notes: Commodities without an observed PPI remain in the input-output system and transmit shocks through the network but do not enter the producer-price regressions as outcomes.

D.6 Additional data

Prices. Consumer-price outcomes are monthly BEA PCE price indexes at the NIPA expenditure-category level. Producer-price outcomes are monthly BLS PPI series matched as described above. Short gaps in individual PPI series are interpolated when adjacent observations are available; longer gaps remain missing.

Concentration. For Appendix F.4, we construct BEA-commodity Herfindahl–Hirschman indexes from firm sales using Economic Census and NETS data mapped through NAICS. The 2025 specification uses the 2022 sales HHI.

Oil and wages. The local projections control for oil and wage movements outside the model. Oil exposure combines each commodity’s petroleum input share with the monthly crude-oil price change. Wage growth is measured at the industry level from BLS data.

E Robustness of the Pass-through Estimates

Each exercise below changes one feature of the baseline local-projection specification while leaving the remaining construction, controls, fixed effects, and inference unchanged.

E.1 The 2018–19 episode

We apply the producer-price specification to the 2018 tariff episode using 2017 input-output matrices and 2017 trade weights. The three exposure measures are constructed as in the 2025 exercise. The treatment window ends with the May 2019 customs observation, the last month before the June 2019 escalation of the List 3 action to 25% on the same goods the 2018 waves had already hit; at the reference horizon its outcome window closes in January 2020, before the pandemic. That gives 10,562 observations on 258 industries. On this sample the criterion (F.6) is minimised in the interior, at $\hat{\rho} = 0.697$, with a 95% confidence set of $[0.238, 1.000]$; evaluating both moments at $\bar{\rho} = 1$ gives $S = 1.38$ against $\chi^2_{2,0.95} = 5.99$, so unlike in 2025 the monopolistic competition limit is not rejected. The panel is drawn to $h = 10$, the deepest horizon whose outcome window still closes before March 2020 from the last treated month; $h = 11$ and $h = 12$ are fitted on the same sample — which is what keeps the horizons comparable — but are neither drawn nor read.

E.2 Pass-through in logs

We rebuild all model-implied exposures under pass-through in logs and re-estimate the baseline local projections.

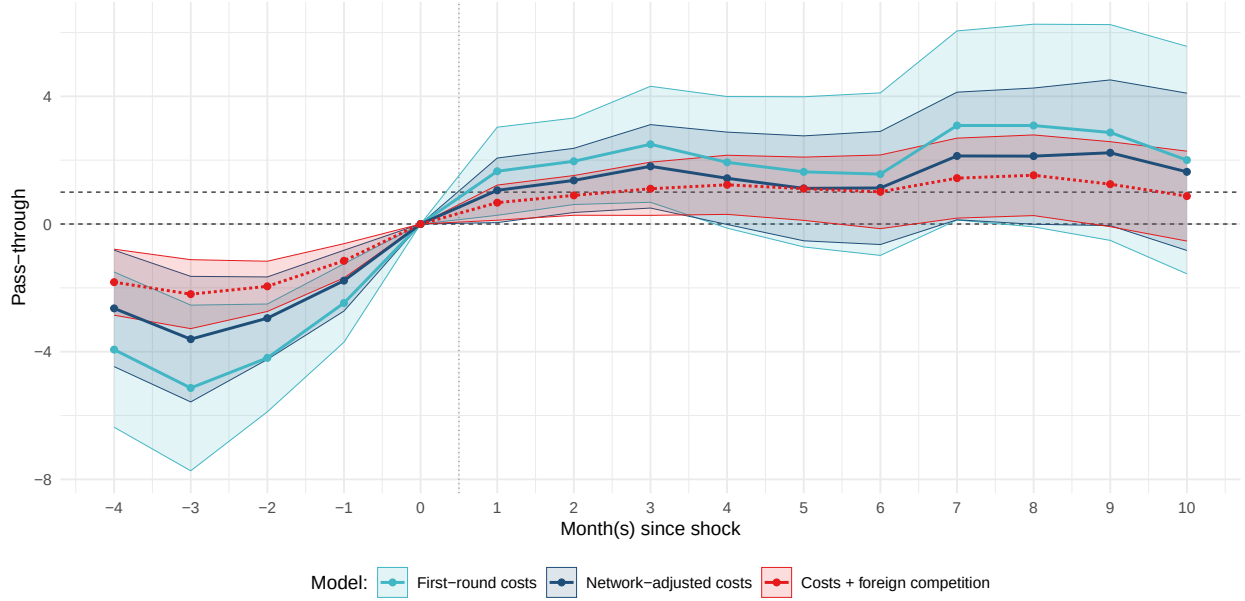
E.3 Sign-symmetry of the tariff-path controls

The alternative specification allows positive and negative statutory tariff changes in the lead and lag controls to enter with separate coefficients. The contemporaneous treatment is unchanged.

E.4 Goods-only PCE

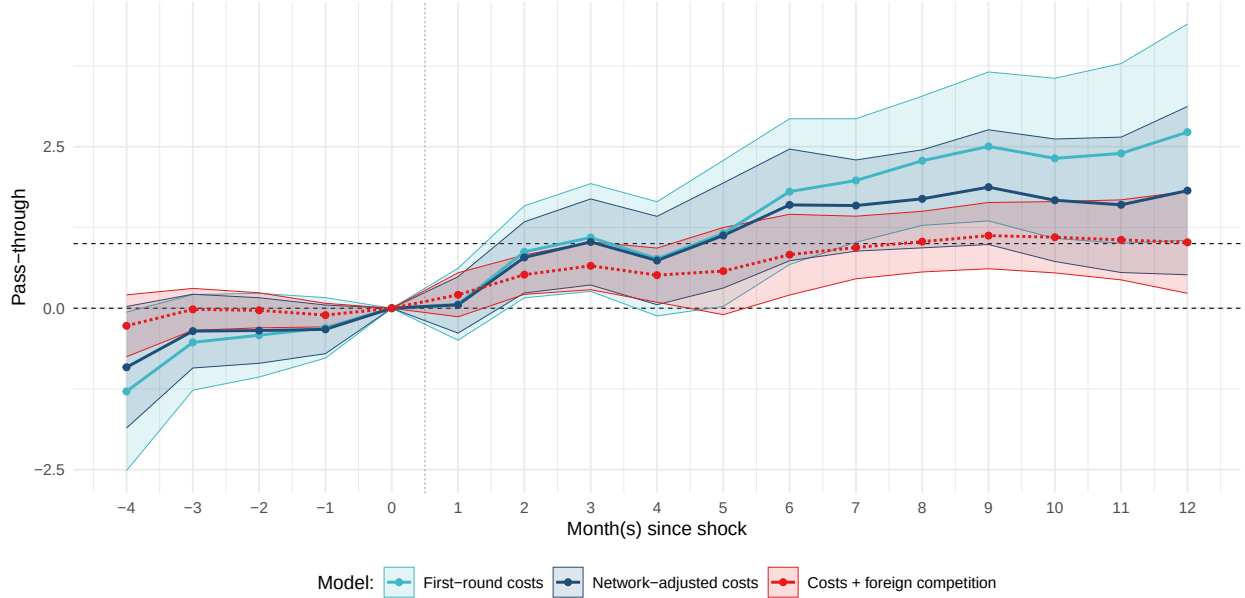
We restrict the PCE panel to goods categories and reconstruct the specification on the retained sample.

Figure E.1: PPI Tariff Pass-through — 2018–19 Episode



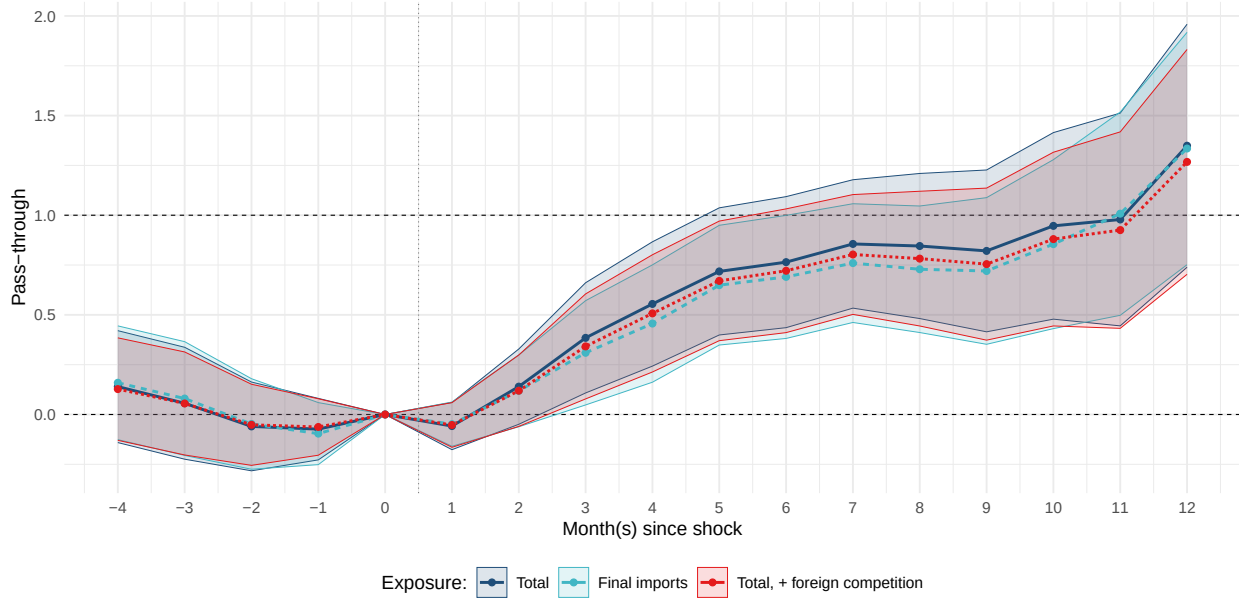
Notes: As Figure 3, applied to the 2018–19 episode using 2017 input-output relationships and import weights. The foreign-competition line is built at $\hat{\rho} = 0.697$, and at the reference horizon it gives $\beta_9 = 1.248$ (0.679) against 2.232 (1.164) for the network-adjusted arm. Horizons are drawn to $h = 10$, the deepest one whose outcome window closes before March 2020. Bands are 95% confidence intervals clustered by industry.

Figure E.2: PPI Tariff Pass-through — Pass-through in Logs



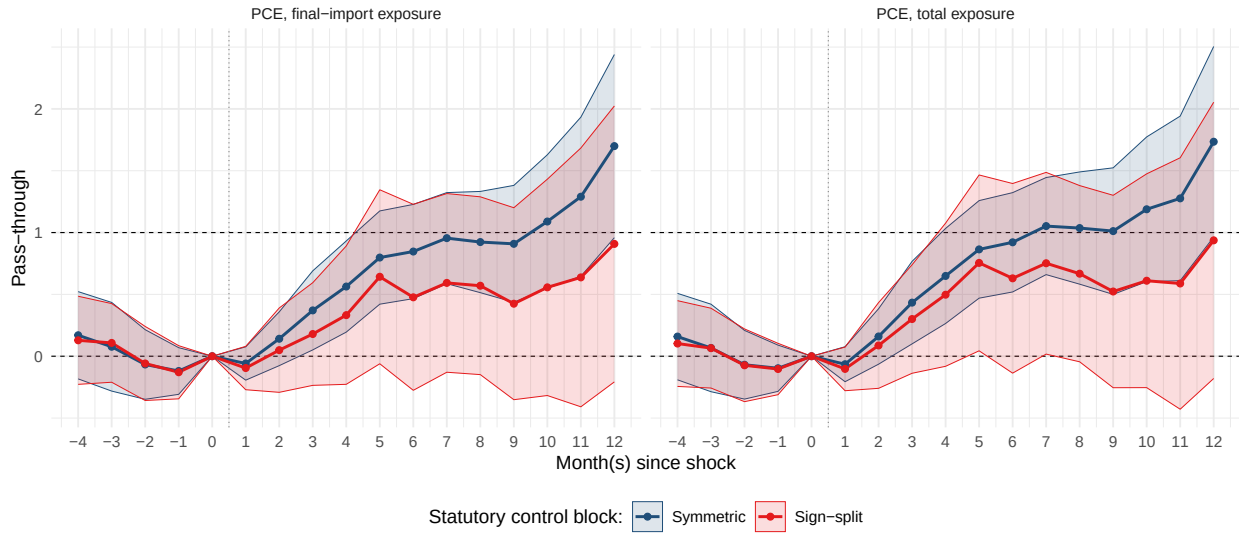
Notes: As Figure 3, with exposures rebuilt under pass-through in logs ($\Upsilon^d = I$).

Figure E.3: PCE Tariff Pass-through — Pass-through in Logs



Notes: As Figure 5, with exposures rebuilt under pass-through in logs ($\Upsilon^d = I$).

Figure E.4: PCE Tariff Pass-through — Sign-symmetric vs. Sign-split Control Block



Notes: As Figure 5, with one panel per exposure measure. The alternative specification enters positive and negative statutory leads and lags separately; the treatment itself is not split. Bands are 95% confidence intervals clustered by item.

Figure E.5: PCE Tariff Pass-through — All Items vs. Goods Only



Confidence band truncated at ± 3 for legibility; estimates and standard errors are unclipped.

Notes: As Figure 5, with one panel per exposure measure. The alternative sample contains goods expenditure categories only. Bands are 95% confidence intervals clustered by item.

E.5 Alternative measures of the tariff shock

We consider two alternatives to the baseline shock construction: the legally enacted tariff schedule as the instrument and the duty-inclusive border unit value as the treatment. Because the enacted and observed statutory series differ mainly in the timing of the short-lived April–May 2025 China tariff increase, these comparisons use doses cumulated over the same window as the price outcome.

Table E.1: Cumulated Pass-through under Alternative Measures of the Tariff Shock

	$h = 6$	$h = 9$	$h = 12$
<i>Panel A. The instrument: statutory observed vs. enacted schedule</i>			
Statutory observed	0.552 (0.267)	0.833 (0.342)	1.247 (0.455)
Enacted schedule	0.678 (0.294)	0.843 (0.359)	0.976 (0.486)
First stage, statutory	0.794	0.797	0.820
First stage, enacted	0.733	0.911	0.891
Hansen J , p -value	0.400	0.948	0.197
<i>Panel B. The treatment: tariff cost index vs. duty-inclusive border price</i>			
Tariff cost index	0.552 (0.267)	0.833 (0.342)	1.247 (0.455)
Duty component of the border price	0.536 (0.258)	0.832 (0.344)	1.263 (0.464)
Duty-inclusive border price	0.683 (0.325)	1.161 (0.459)	1.347 (0.432)
First stage, cost index	0.794	0.797	0.820
First stage, duty component	0.818	0.798	0.810
First stage, ex-tariff component	-0.176	-0.226	-0.050
First stage, duty-inclusive	0.642	0.572	0.759
Items	273	272	272

Notes: One observation per producer-price item. The dependent variable is the log price change from December 2024 to the indicated horizon and the treatment is the corresponding cumulated tariff or border-price dose. Panel A changes the instrument; Panel B changes the treatment. Exposure is constructed on Detail BEA commodities using fixed 2024 source-region weights. Heteroskedasticity-robust standard errors are reported.

F Robustness of the Strategic-Complementarity Estimate

The baseline estimates impose a common conditional own-cost pass-through parameter $\bar{\rho}$ and abstract from within-industry covariance between firm-level pass-through and imported-input exposure. We consider three extensions: joint estimation of $\bar{\rho}$ and a common covariance wedge, estimation of $\bar{\rho}$ conditional on that wedge, and a specification in which ρ varies with market concentration.

F.1 Estimation

Let

$$y_{i,t} = \Delta_{h_{\text{ref}}} \log p_{i,t+h_{\text{ref}}}$$

and let $\tilde{x}$ denote the residual from projecting x on the fixed effects and controls in equation (50). For parameter vector θ , define

$$u_{i,t}(\theta) = \tilde{y}_{i,t} - \widetilde{\Delta \log p_{i,t}}^M(\theta). \quad (\text{F.1})$$

The instruments are the statutory model-implied response and its parameter derivatives,

$$z_{0,i,t}(\theta) = \widetilde{\Delta \log p_{i,t}}^{M,\text{stat}}(\theta), \quad (\text{F.2})$$

$$z_{k,i,t}(\theta) = \left(\frac{\partial \widetilde{\Delta \log p_{i,t}}^{M,\text{stat}}(\theta)}{\partial \theta_k} \right), \quad k = 1, \dots, p, \quad (\text{F.3})$$

with moments

$$g_k(\theta) = \sum_{i,t} z_{k,i,t}(\theta) u_{i,t}(\theta).$$

We use continuously updated GMM with an item-clustered covariance matrix,

$$\widehat{\mathbf{V}}(\theta) = \sum_c \mathbf{G}_c(\theta) \mathbf{G}_c(\theta)', \quad (\text{F.4})$$

$$\mathbf{G}_c(\theta) = \sum_{(i,t) \in c} \mathbf{z}_{i,t}(\theta) u_{i,t}(\theta), \quad (\text{F.5})$$

and criterion

$$S(\theta) = \mathbf{g}(\theta)' \widehat{\mathbf{V}}(\theta)^{-1} \mathbf{g}(\theta). \quad (\text{F.6})$$

Confidence sets are obtained by test inversion — $\mathcal{S}_{0.95}$ collects the parameter values the

criterion does not reject,

$$\mathcal{S}_{0.95} = \{\theta : S(\theta) \leq \chi_{K,0.95}^2\}, \quad (\text{F.7})$$

where K is the number of moments. For multi-parameter specifications, reported intervals are projections of this set. No resampling is involved: the data enter once, and it is the candidate value of θ that is varied. This is deliberate rather than a matter of convenience. At any fixed θ_0 the moments $\mathbf{g}(\theta_0)$ are sample averages, so their distribution does not depend on how strongly θ is identified and the χ_K^2 cutoff is valid whether identification is strong or weak. A standard error read off the curvature of S at $\hat{\theta}$ would not be, and neither would a bootstrap of $\hat{\theta}$: both require $\hat{\theta}$ itself to be well behaved, which is what fails when the criterion is flat over stretches of the parameter space. The cost is intervals that are wider and, in directions the data do not speak to, unbounded. The two-parameter arms search $\bar{\rho}$ on a 0.02 grid over $[0.10, 1]$. The floor keeps the implied ρ_j away from the corner in which the operator inverted in (45) approaches singularity and the predicted cross-section and its derivatives diverge together; how close an estimate sits to it is reported alongside the estimate rather than used to move one.

F.2 Joint estimation of $\bar{\rho}$ and the covariance wedge

Under

$$\mathbb{C}^d = \mathbf{0}, \quad \mathbb{C}^* = \mathbf{\Gamma} \mathbf{\Omega}^*,$$

and common parameters $\rho_i = \bar{\rho}$ and $\gamma_i = \bar{\gamma}$, producer exposure becomes

$$\mathbf{Z}(\bar{\rho}, \bar{\gamma}) = \left[\mathbf{I} - (1 - \bar{\rho})(\mathbf{I} - \mathbf{S}_F) - \bar{\rho} \mathbf{\Upsilon}^d \mathbf{\Omega}^d \right]^{-1} \left[(\bar{\rho} + \bar{\gamma}) \mathbf{\Upsilon}^d \mathbf{\Omega}^* + (1 - \bar{\rho}) \mathbf{S}_F \right]. \quad (\text{F.8})$$

We estimate $(\bar{\rho}, \bar{\gamma})$ using the scale moment and the two parameter-derivative moments. Since $\bar{\gamma} = \rho^* - \bar{\rho}$, where ρ^* is an exposure-weighted average of firm-level pass-throughs, its structural range is

$$-\bar{\rho} \leq \bar{\gamma} \leq 1 - \bar{\rho}. \quad (\text{F.9})$$

We report but do not impose this restriction, searching $\bar{\gamma}$ over $[-4, 4]$.

The joint estimate, $\hat{\bar{\rho}} = 0.719$, is close to the baseline estimate. The covariance wedge is weakly identified: its confidence set contains zero and its unconstrained point estimate lies outside its structural range. We therefore interpret this exercise as a robustness check on $\bar{\rho}$, rather than as a separate estimate of $\bar{\gamma}$.

Table F.1: Joint Estimation of $\bar{\rho}$ and the Covariance Wedge

Parameter	Estimate	95% projected set
$\bar{\rho}$	0.719	[0.403, ≥ 1.000]
$\bar{\gamma}$	0.716	[-0.88, 2.64]
$\hat{\bar{\rho}} + \hat{\bar{\gamma}}$, admissible in $[0, 1]$	1.435	
J (1 df), p	0.01	0.925
$S(\hat{\bar{\rho}}, 0)$, p (3 df)	2.76	0.430

Notes: $\bar{\rho}$ and $\bar{\gamma}$ are estimated jointly using the scale moment and their two derivative moments, with exposure (F.8). Confidence sets are projections of the non-rejected region (F.7) and are conservative; an end marked $\geq$ sits on the edge of the searched region, $\bar{\rho} \in [0.10, 1]$ and $\bar{\gamma} \in [-4, 4]$. The implied exposure-weighted pass-through is $\bar{\rho} + \bar{\gamma}$, whose structural range is $[0, 1]$ and is not imposed. J tests the single over-identifying restriction against χ^2_1 , and $S(\hat{\bar{\rho}}, 0)$ — the criterion with the wedge switched off at the jointly estimated $\hat{\bar{\rho}}$ — is read against $\chi^2_{3,0.95} = 7.81$. Producer prices, 2025 episode, pass-through in levels, $h = 9$.

F.3 $\bar{\rho}$ conditional on the covariance wedge

We next fix $\bar{\gamma}$ at alternative values and re-estimate $\bar{\rho}$ using the same moment system.

Table F.2: $\hat{\bar{\rho}}$ under Imposed Values of the Covariance Wedge

Imposed $\bar{\gamma}$	$\hat{\bar{\rho}}$	S at $\hat{\bar{\rho}}$	95% set for $\bar{\rho}$	$\hat{\bar{\rho}} + \bar{\gamma}$
-1.00	—	8.59	(empty)	— — —
-0.50	0.600	5.08	[0.46, 0.76]	0.10
-0.20	0.620	3.16	[0.44, 0.84]	0.42
0.00	0.640	2.02	[0.42, 0.88]	0.64
0.50	0.680	0.20	[0.44, 0.96]	1.18
1.00	0.760	0.30	[0.52, ≥ 1.00]	1.76

Notes: $\bar{\gamma}$ is fixed at each listed value and $\bar{\rho}$ is re-estimated on a 0.02 grid using the same three moments as in Table F.1. The column S at $\hat{\bar{\rho}}$ is the criterion with $\bar{\rho}$ concentrated out, read against $\chi^2_{3,0.95} = 7.81$, so it tests the imposed wedge; the 95% set for $\bar{\rho}$ collects the values that same criterion does not reject at the given wedge. The column $\hat{\bar{\rho}} + \bar{\gamma}$ reports the implied exposure-weighted pass-through, admissible in $[0, 1]$ by (F.9). Producer prices, 2025 episode, pass-through in levels, $h = 9$.

Across the reported values of $\bar{\gamma}$, the estimate of $\bar{\rho}$ remains close to the headline estimate. In particular, admissible nonpositive covariance wedges imply somewhat lower $\bar{\rho}$, strengthening rather than eliminating the foreign-competition channel.

F.4 Allowing ρ to vary with market concentration

We finally allow ρ to vary across commodities with effective market concentration:

$$\rho_j = 1 - \alpha_0 - \alpha_1(1 - s_{Fj})HHI_j. \quad (\text{F.10})$$

Here HHI_j is the sales Herfindahl index and $(1 - s_{Fj})HHI_j$ measures domestic concentration after accounting for the foreign market share.

We estimate (α_0, α_1) using the scale moment and the two corresponding derivative moments.

Table F.3: ρ Parameterized by Effective Concentration

Parameter	Estimate	95% projected set
α_0	0.295	[0.00, 0.60]
α_1	0.899	[-0.90, 1.35]
Implied ρ_j , 10th pct	0.498	
median	0.652	
90th pct	0.690	
J (1 df), p	1.28	0.258
S at $\alpha_1 = 0$ (one common ρ), p (3 df)	1.93	0.587
S at $\alpha_1 = 1$ (microfoundation), p (3 df)	1.47	0.688
Largest α_1 keeping every $\rho_j \geq 0.10$	0.960	
Active commodities	222	
with $(1 - s_F)HHI > 0$	222	

Notes: Commodity-level pass-through is parameterized according to equation (F.10). HHI_j is the 2022 sales Herfindahl index and s_{Fj} the foreign sales share, so the gradient α_1 is identified off the commodities with a strictly positive effective index. Confidence sets are projections of the non-rejected region (F.7) and are conservative. J tests the single over-identifying restriction against χ^2_1 ; the two S rows are the profiled criterion with the stated restriction imposed, read against $\chi^2_{3,0.95} = 7.81$. The row *Largest α_1 keeping every $\rho_j \geq 0.10$* is arithmetic rather than statistical, evaluated at $\hat{\alpha}_0$, and records how close the estimate sits to the grid floor. Producer prices, 2025 episode, pass-through in levels, $h = 9$.

The estimates, $\hat{\alpha}_0 = 0.295$ and $\hat{\alpha}_1 = 0.899$, imply a median commodity-level ρ_j of 0.652, close to the common- ρ estimate. The estimated concentration gradient has the sign predicted by the model, but its confidence set includes zero. We therefore view this specification as consistent with, rather than independently identifying, the model's market-structure mechanism.